



Predicting exchange rate with commodity prices: New evidence from Westerlund and Narayan (2015) estimator with structural breaks and asymmetries

Afees A. Salisu^{a,b,*}, Wasiu Adekunle^c, Wasiu A. Alimi^c, Zachariah Emmanuel^c

^a Department for Management of Science and Technology Development, Ton Duc Thang University, Ho Chi Minh City, Viet Nam

^b Faculty of Business Administration, Ton Duc Thang University, Ho Chi Minh City, Viet Nam

^c Centre for Econometric & Allied Research, University of Ibadan, Nigeria

ARTICLE INFO

Keywords:

Exchange rate

Commodity prices

Forecast evaluation

Asymmetry

Structural break/JEL classification:

F31

F37

Q02

ABSTRACT

In this paper, we offer new evidence on the predictability of exchange rate with commodity prices using the Westerlund and Narayan (2015) estimator. The estimator allows us to control for any possible bias in the estimation process due to the presence of persistence, endogeneity and conditional heteroscedasticity effects in our predictors. We further account for the role of asymmetries and structural breaks both for the in-sample and out-of-sample forecasts. In particular, we evaluate whether such considerations matter for the forecast performance of the predictive model for exchange rate. Monthly data of five major tradable currency pairs in the world and disaggregated commodity price indices over the period of 1960–2017 are utilized. We find significant improvements in both the in-sample and out-of-sample forecast performance of the predictive model for exchange rate using this estimator and particularly when asymmetries and structural breaks are accommodated. While the results are robust to multiple currency pairs, commodity price indices, breaks and forecast horizons, however, the choice of data frequency has implications on the forecast outcome. Essentially, the forecast accuracy of the proposed predictive model seems to diminish with lower data frequencies.

1. Introduction

The relationship between commodity prices and exchange rate has been well examined by a substantial body of the literature and most of them lend support to the predictability of commodity prices in the exchange rate model (see for example in the last two decades or so, [Smith, 1999](#); [Chen et al., 2010](#); [Trezzi, 2014](#); [Makin, 2013](#); [Choudhri and Schembri, 2013](#); [Arezki et al., 2014](#); [Dauvin, 2014](#); [Bodart et al., 2015](#); [Ignatieva and Ponomareva, 2016](#); [Medina and Soto, 2016](#); [Harris and Shen, 2017](#); [Yang et al., 2017](#); [Lu et al., 2017](#)). However, a common limitation of most of these studies is that their analyses are restricted to in-sample predictability and the decisive test of any predictive model is its out-of-sample performance ([Campbell, 2008](#)). The only few exceptions in this regard are [Chen et al. \(2010\)](#); [Ferraro et al. \(2015\)](#); [Chen et al. \(2016\)](#); and [Kohlscheen et al. \(2016\)](#) which attempt to forecast exchange rate using commodity prices both for the in-sample and out-of-sample data. All but [Chen et al. \(2010\)](#) largely support the significant

role of commodity prices in predicting exchange rate both for the in-sample and out-of-sample forecasts.

In this study, we advance the literature on the predictability of commodity prices in the exchange rate model by offering the following innovations. First, we evaluate the in-sample and out-of-sample predictability of commodity price-exchange rate nexus. All the known studies on the subject are limited to the in-sample analyses. The need to conduct a distinct analysis for out-of-sample forecast is usually motivated by the conclusion of [Bossaerts and Hillion \(1999\)](#), [Goyal and Welch \(2003\)](#), [Brennan and Xia \(2005\)](#), [Butler et al. \(2005\)](#), [Ang and Bekaert \(2007\)](#), [Ang et al. \(2007\)](#) and [Chen et al. \(2010\)](#). These studies indicate that return predictability is predominantly in-sample; and it is not robust to out-of-sample evaluations. While there are recent attempts at predictability analyses as previously noted (see footnote 1), our study is the first to consider useful generalization of the out-of-sample predictability of commodity prices in the forecast of exchange rate, covering an array of major commodity prices and exchange rates.¹ The

* Corresponding author. Department for Management of Science and Technology Development, Ton Duc Thang University, Ho Chi Minh City, Viet Nam.

E-mail address: afees.adebare.salisu@tdtu.edu.vn (A.A. Salisu).

¹ Nonetheless, there are studies on commodity prices although not from the perspective of exchange rate forecasting. Examples include [Nazlioglu \(2011\)](#), [Chkili et al. \(2014\)](#), [Rafiq and Bloch \(2016\)](#), [Salisu et al. \(2018a\)](#), [Ubilava \(2017\)](#), [Bouri et al. \(2018\)](#), [Apergis and Vouzavalis \(2018\)](#), [Raza et al. \(2018\)](#) and [Shahzad et al. \(2018\)](#).

only exceptions are [Chen et al. \(2010\)](#) and [Ferraro et al. \(2015\)](#) both of which also test the predictability of commodity prices in exchange rate. However, we depart from these studies in terms of the choice of methodology, exchange rates and commodity prices. Also, the two related studies substantially lend support to the [Mese and Rogoff \(1983\)](#) puzzle which states that there is hardly any economic model that can beat the random walk model when forecasting exchange rate. In contrast, as you will find later in the write-up, we are able to establish a strong connection between the two variables of interest if certain considerations are captured in the model.

Our second motivation relates to the inclusion of asymmetries in the predictive model. The intention is to test whether such consideration matters for forecasting of exchange rate especially as such financial series exhibit nonlinearity by their nature; that is, exchange rate appreciation and depreciation. Recent studies involving commodity prices, although not from the perspective of exchange rate forecasting, have argued for the role of asymmetries (see for example, [Herrera et al., 2010](#); [Kilian and Vigfusson, 2011](#); [Nazlioglu, 2011](#); [Chkili et al., 2014](#); [Narayan and Gupta, 2015](#); [Bahmani-Oskooee and Saha, 2016](#); [Rafiq and Bloch, 2016](#); [Ubilava, 2017](#); [Valadkhani and Smyth, 2017](#); [Apergis and Vouzavalis, 2018](#); [Bouri et al., 2018](#); [Raza et al., 2018](#); [Salisu et al., 2018a](#); [Shahzad et al., 2018](#)). In addition to the fact that most of these studies are restricted to in-sample forecast analyses, there is no study yet, to the best of our knowledge that attempts to validate such consideration when forecasting exchange rate particularly for the out-of-sample forecast. The only exception is the study of [Ferraro et al. \(2015\)](#) however the consideration was restricted to a single commodity price (which is oil price) and a single exchange rate (which is the Canadian-US dollar exchange rate). Therefore, the results of [Ferraro et al. \(2015\)](#) may not be valid for possible generalization.

The third contribution involves the inclusion of structural breaks in the predictive model for commodity price-exchange rate nexus. In other words, we test whether accounting for structural breaks matters for the predictability of exchange rate with commodity prices. The underlying motivation for this analysis is derived from the findings of [Paye and Timmermann \(2006\)](#), [Rapach and Wohar \(2006, 2008\)](#), [Pesaran and Timmermann \(2007\)](#), [Giacomini and Rossi \(2009\)](#) and [Pesaran et al. \(2013\)](#). These studies argue that detecting and accounting for structural break if it exists may matter for both in-sample and out-of-sample forecasts particularly for high frequency series. In particular, [Pesaran and Timmermann \(2007\)](#) and [Pesaran et al. \(2013\)](#) suggest that accounting for structural break, even when modeling with time varying parameter regressions, makes forecasts more precise. Like the asymmetry case, the significance of structural breaks in the predictability of exchange rate with commodity prices is evaluated for both the in-sample and out-of-sample forecast.

The fourth motivation involves the need to reflect the special features fundamental to the predictor series, commodity prices and the resulting implication in the estimation and forecasting process. Essentially, we suspect that there may be need to account for any salient feature inherent in the predictor such as endogeneity, persistence and conditional heteroscedasticity effects. The issue of endogeneity is not unexpected since, for convenience and to isolate the predictability of commodity prices in exchange rate model, a single-factor predictor is considered. Therefore, it is not out of place to suspect such bias where there is correlation between the predictor and the regression error. Given the implication of the bias on forecast outcomes ([Lewellen, 2004](#); [Westerlund and Narayan, 2012, 2015](#)), resolving it becomes crucial. Also, most economic series including commodity prices tend to exhibit some persistence effect due to their susceptibility to shocks and therefore accounting for same may improve the forecast results. Meanwhile, while analyzing the predictability of stock returns, [Lewellen \(2004\)](#) demonstrates that ignoring endogeneity and persistence effects may

affect the forecast performance of a predictive model.² Lastly, we utilize high frequency and therefore the issue of conditional heteroscedasticity effect has to be dealt with in the estimation process and the works of [Westerlund and Narayan \(2012, 2015\)](#) find that exploiting the information contained in the heteroscedasticity effect may enhance forecast precision.³ Therefore, our choice of methodology relies on the procedure of [Lewellen \(2004\)](#) and [Westerlund and Narayan \(2012, 2015\)](#) that accommodates some important features inherent in most economic and financial predictors as those highlighted.

Finally, in order to test for the superiority of the commodity price-based predictive model for forecasting exchange rate over the time series model, we compare their forecast performance with the historical average model. Thereafter, the single-factor commodity price-based predictive model is compared with other (extended) variants involving additional parameters for asymmetries and structural breaks. This is to validate our hypothesis that accounting for these parameters will enhance the forecast accuracy of the commodity-based predictive model for exchange rate.

For completeness, we offer several robustness checks as follows. First, we consider different exchange rates based on the major tradable currency pairs in the world involving the currencies of Australia, Canada, Japan, the United Kingdom and the Euro Area with US Dollar being the reference currency for the others. [Salisu et al. \(2018b\)](#) note the major tradable currency pairs tend to react differently to shocks and therefore results from a single currency pair cannot be used to generalize. Secondly, we utilize data on different commodity-specific indices namely the agricultural price index, crude oil price index, energy price index, price of gold, and industrial metal price index and their forecasts are singly evaluated. This is motivated by the study of [Narayan and Sharma \(2011\)](#) which find that oil price affects returns of firms differently depending on their sectoral location. Similar evidence is reported in the studies of [Narayan et al. \(2018\)](#) and [Salisu et al. \(2019\)](#). Drawing from this finding, we also hypothesize that exchange rate may respond differently to commodity prices depending on the preference of the countries in question for the selected commodities. Thirdly, we control for other factors that predict exchange rate and thereafter we test whether our original conclusion will remain unchanged as regards the predictability of commodity prices in exchange rate forecasting. This idea is also inspired by the study of [Narayan et al. \(2016a\)](#) and [Salisu et al. \(2019\)](#) suggesting that the inclusion of macroeconomic variables in the oil-stock predictability may enhance its forecast accuracy. Fourth, we also test whether our results are robust to alternative data frequencies. This consideration is motivated by the studies of [Ferraro et al. \(2015\)](#), [Narayan and Liu \(2015\)](#), [Narayan and Sharma \(2015\)](#), [Salisu and Adeleke \(2016\)](#) and [Salisu et al. \(2016\)](#) indicating that the nature of data frequency may matter in forecast analyses. The analyses for all the robustness tests are conducted for both the in-sample and out-of-sample periods.

Our results support the consideration of asymmetries and structural breaks when forecasting exchange rates with commodity prices. While the results are robust to multiple currency pairs, commodity price indices, breaks and forecast horizons, however, the choice of data frequency has implications on the forecast outcome. Essentially, the forecast accuracy of the proposed predictive model seems to diminish with lower data frequencies.

The remaining part of this paper is structured as follows: Section 2

² Similar conclusions have been reached by studies focusing on inflation forecasting (see [Salisu and Isah, 2018](#); [Salisu et al., 2018a](#)) and stock returns forecasting (see [Bannigidadmath and Narayan, 2015](#); [Narayan and Bannigidadmath, 2015](#); [Narayan and Gupta, 2015](#); [Phan et al., 2015](#); [Westerlund and Narayan, 2015](#); [Narayan et al., 2016a](#); [Devpura et al., 2018](#); and [Salisu et al., 2019](#)). This approach was first developed by [Lewellen \(2004\)](#) and extended by [Westerlund and Narayan \(2012, 2015\)](#).

³ This claim has been empirically validated by the same studies listed under footnote 2.

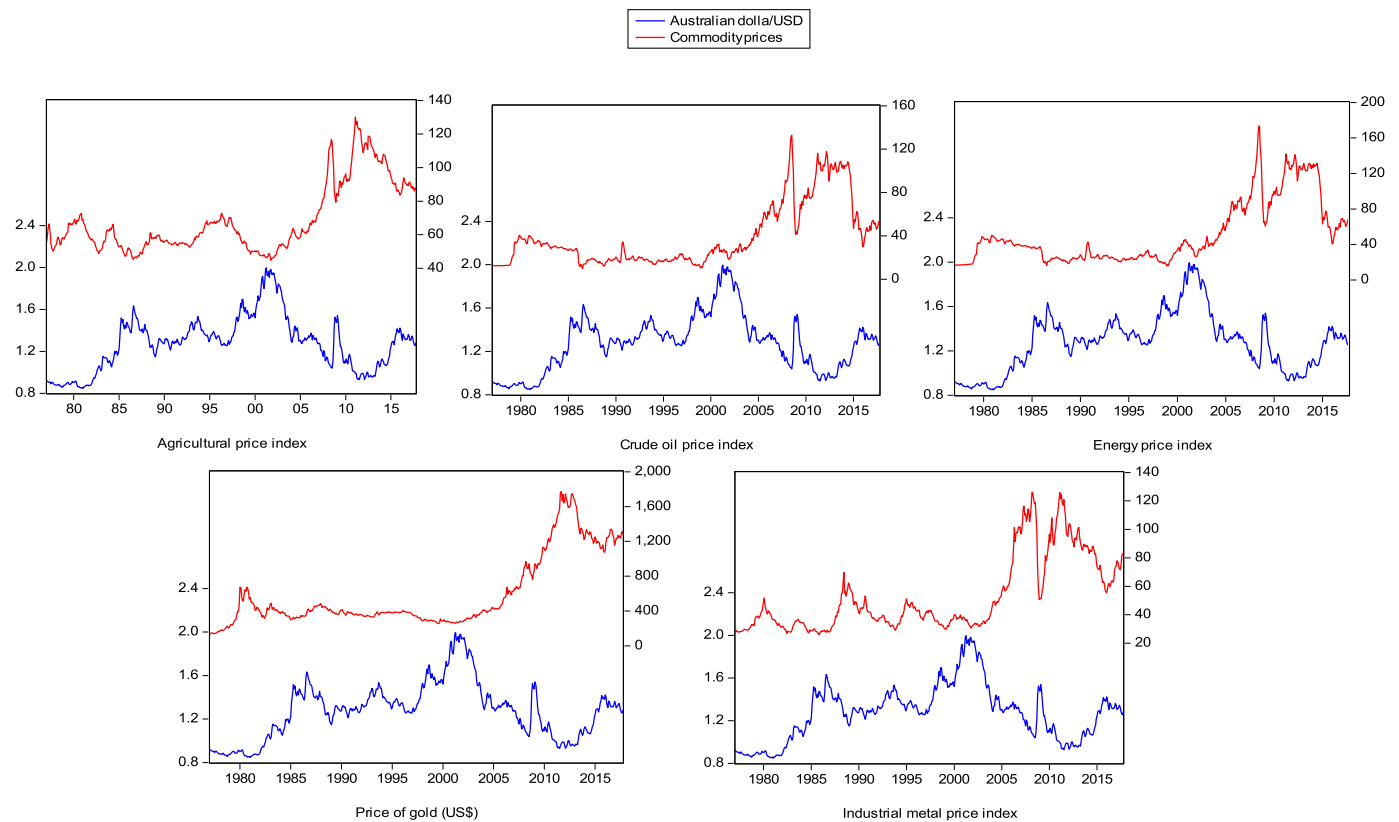


Fig. 1. Commodity-specific price indices and Australian-US dollar exchange rate.

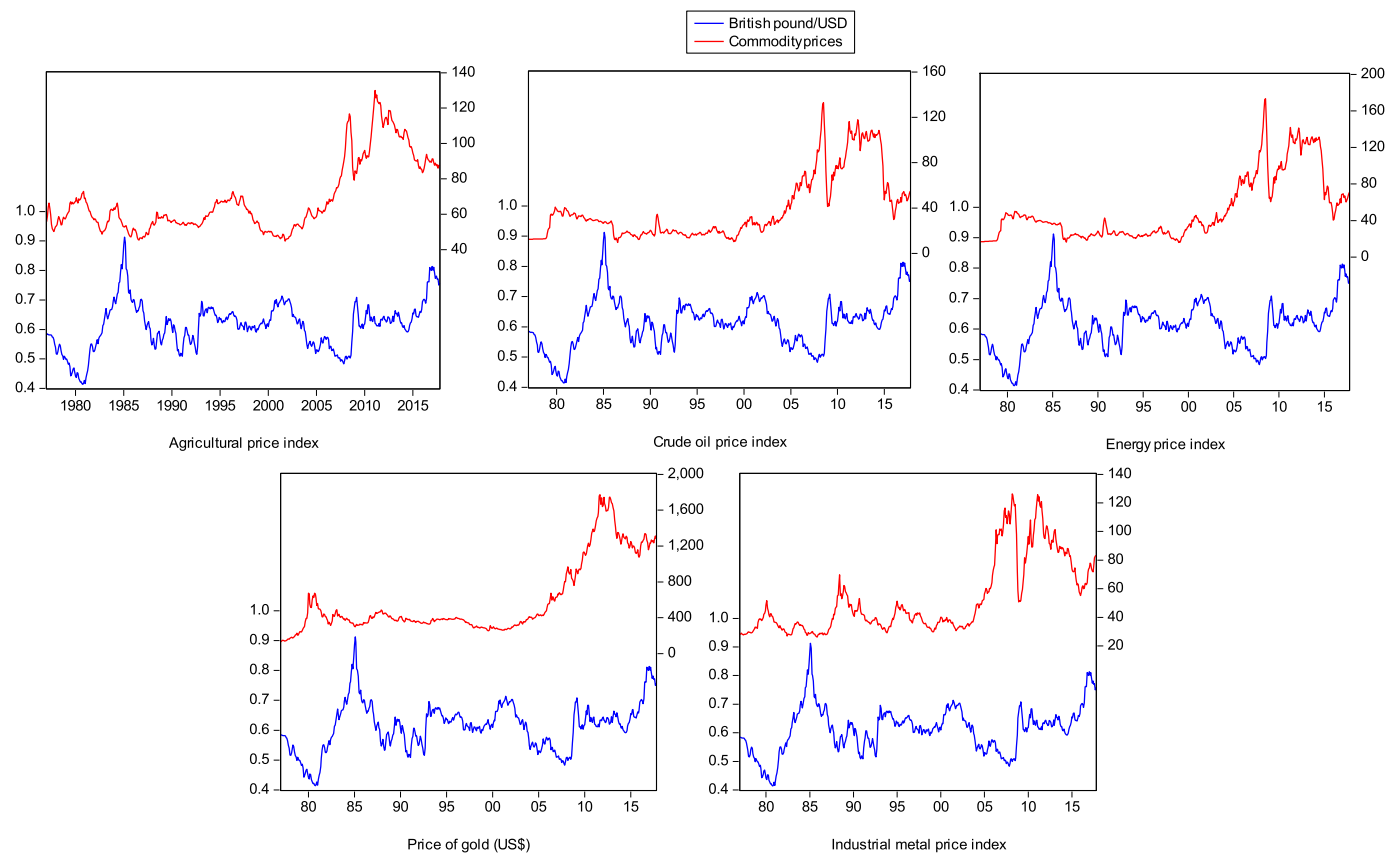


Fig. 2. Commodity-specific price indices and British pound-US dollar exchange rate.

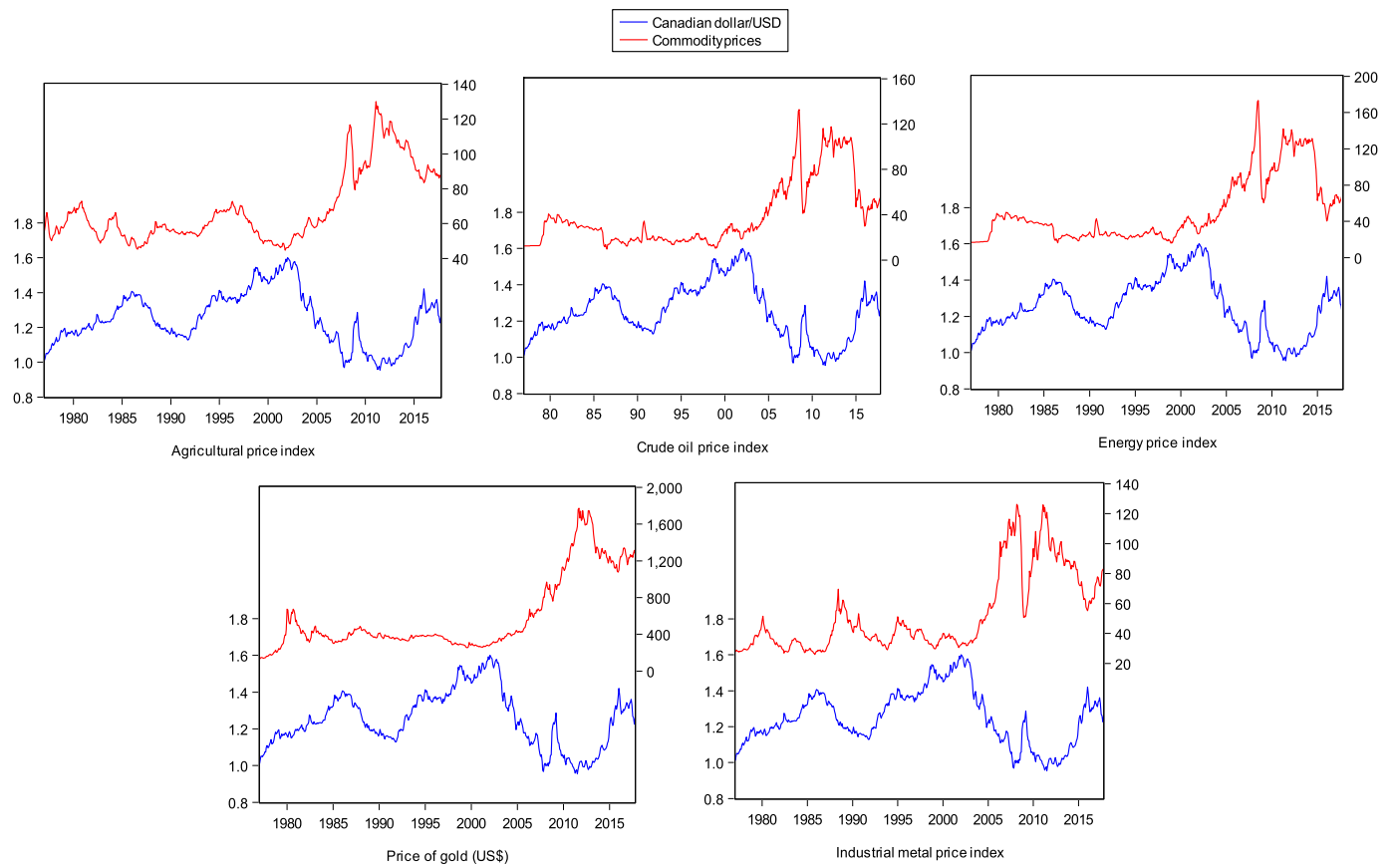


Fig. 3. Commodity-specific price indices and Canadian-US dollar exchange rate.

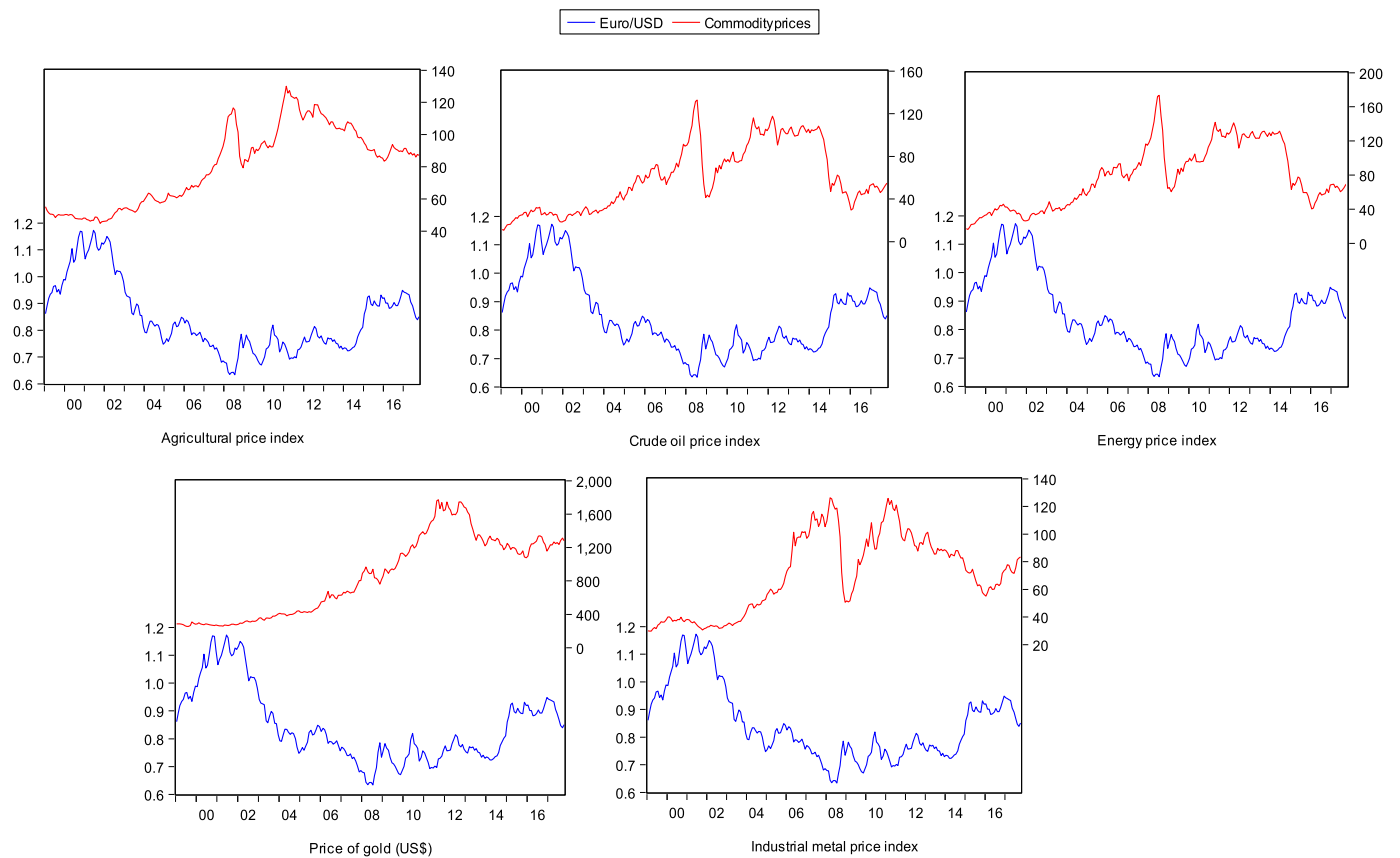


Fig. 4. Commodity-specific price indices and Euro-US dollar exchange rate.

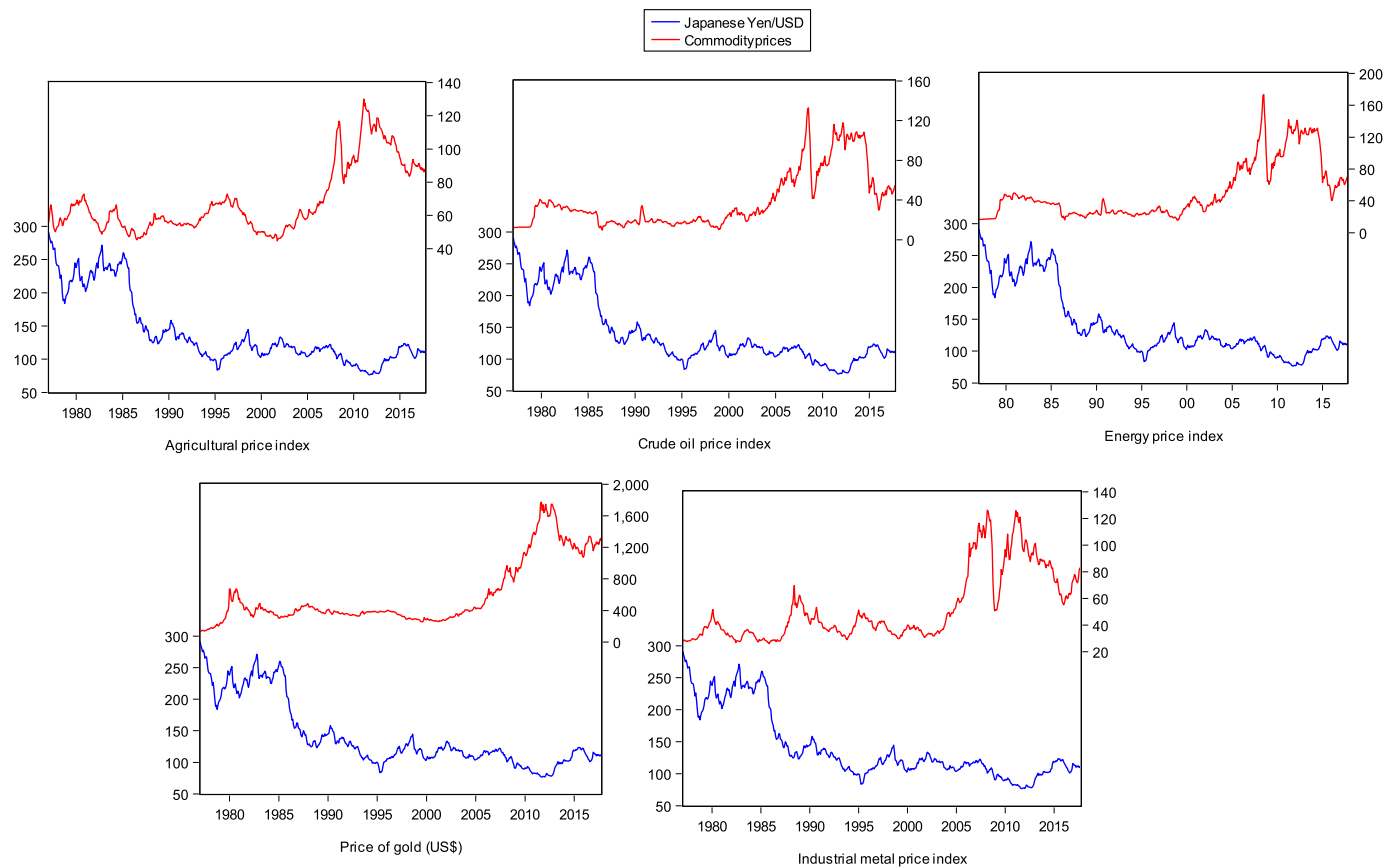


Fig. 5. Commodity-specific price indices and Japanese Yen-US dollar exchange rate.

Table 1

Data scope and sample size description.

Variables	Start period	End Period	No. of Observations
All the Commodity prices [AG; CR; EN; GO; ID]	January, 1977	October, 2017	490
Euro/US \$ exchange rate [EUR]	January, 1999	October, 2017	226
Other most traded currency pairs [AUD; CAD; BRP; JPY]	January, 1977	October, 2017	490

Note: Data on commodity price indices were obtained from the World Bank Database, while the data on exchange rates were sourced from the IMF-IFS. Commodity prices include agricultural price index (AG), crude oil price index (CR), energy price index (EN), price of gold (GO), and industrial metal price index (ID). Other most traded currency pairs include exchange rates of Australian (AUD), Canadian dollar (CAD), British pound (BRP) and Japanese Yen (JPY) relative to the US dollar.

discusses relevant theoretical issues. Section 3 provides the model set up including the forecast performance measures. Section 4 deals with data and preliminary analysis while empirical result is presented in Section 5. Section 6 concludes the paper.

2. Theoretical relationship between commodity prices and exchange rate

The direction of relationship between commodity prices and exchange rate is usually premised on the nature of the economy in question. For instance, commodity-exporting currencies are expected to appreciate in response to rising commodity prices and through the supply effect. In other words, increases in demand for commodity exports owing to improved global economic activities probably from favourable investment climate may put pressure on global supply and by extension leading to an increase in commodity prices. And since these commodities are usually traded in foreign currencies particularly US dollar, the increases in the commodity prices will secure more dollar receipts (revenue) for the exporting countries thereby increasing the supply of the currency in the domestic economy relative to demand. This by implication will lead to the appreciation of the domestic

currency relative to US dollar. This can be described as the supply effect of commodity-exchange rate nexus. Conversely, net importers of commodities are expected to experience a depreciation of their currencies relative to the US dollar for example through the demand effect. In other words, higher commodity prices will increase the demand for US dollar by the importing countries in order to settle higher import bills on commodities. This will lead to an appreciation of US dollar relative to the domestic currency of the importing country in question.

The concept of Dutch Disease is also used to explain the nexus between the two variables of interest. In the view of Frankel (2010), the phenomenon of Dutch Disease arises when a strong, but perhaps temporary, upward swing in the world price of export commodity causes, among others, a large real appreciation of the domestic currency. The Dutch disease can arise from sources other than a rise in the commodity price. Others arise from commodity booms due to the discovery of new deposits or some other expansion in supply, leading to a trade surplus via exports or a capital account surplus via inward investment to develop the new resource. In all cases, the result is real appreciation and a shift of resources into non-tradables away from (non-booming) tradables. The appreciation takes the form of a nominal appreciation if the exchange rate is flexible and inflation if the exchange rate is fixed

Table 2
Preliminary analysis.

	Summary Statistics for the Variables					Autocorrelation Test for Predictors			Heteroscedasticity Test for Predictors		
	Mean	Std.	Skw	Kurt	J-B stat	k = 4	k = 8	k = 12	K = 4	K = 8	K = 12
Full-Sample Period: Commodity Prices											
CR	28.790	29.534	1.450	4.381	298.594***	31.526***	39.225***	44.664***	3.714***	1.852*	1.226
GO	435.576	422.057	1.425	4.224	278.465***	52.115***	74.599***	110.45***	12.676***	7.215***	5.691***
AG	56.994	25.468	0.560	2.975	36.383***	152.27***	153.71***	157.63***	29.993***	16.994***	11.632***
EN	37.751	36.906	1.340	4.118	244.051***	33.359***	39.055***	47.541***	1.929*	0.959	0.641
ID	42.449	27.413	1.237	3.727	192.558***	76.483***	86.149***	91.374***	33.983***	17.818***	12.004***
Full-Sample Period: Exchange Rates											
EUR	0.846	0.131	0.818	2.829	25.517***	20.345***	21.527***	23.471**	3.645***	2.251**	1.897**
AUD	1.151	0.293	0.591	2.692	43.237***	87.267***	88.206***	96.161***	311.528***	156.307***	107.100***
BRP	0.549	0.122	−0.097	2.184	20.356	64.920***	70.833***	73.584***	5.236***	2.991***	1.983**
CAD	1.187	0.163	0.587	2.366	51.582***	59.401***	64.589***	79.777***	6.669***	8.328***	7.884***
JPY	199.548	102.317	0.514	1.647	83.457***	67.258***	81.201***	103.13***	12.616***	6.889***	4.785***

Note: CR, GO, AG, EN, and ID, are respectively, crude oil price index, price of gold, agricultural price index, energy price index, and industrial metal price index. EUR, AUD, BRP, CAD, and JPY, respectively, the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. Std is standard deviation, Skw is skewness, Kurt is Kurtosis, and J-B stands for Jarque-Bera. For autocorrelation and heteroscedasticity tests, the reported values are the Ljung-Box test Q-statistics for the former and the ARCH-LM test F-statistics in the case of the latter. We consider three different lag lengths (k) of 4, 8, and 12 for robustness. The null hypothesis for the autocorrelation test is that there is no serial correlation, while the null for the ARCH-LM test is that there is no conditional heteroscedasticity. *** and ** imply the rejection of the null hypothesis in both cases at 1% and 5% levels of significance, respectively.

Table 3
Results of Unit root tests.

Variable	ADF Test			NL Test			
	Level	FD	I(d)	Level	FD	I(d)	Break dates
AUD	−1.977	−16.831***	I (1)	−3.004	−13.969***	I (1)	2003M05; 2009M04
BRP	−2.781	−19.352***	I (1)	−0.479	−14.993***	I (1)	1985M04; 1992M09; 1998M07; 2008M09.
CAD	−2.051	−19.724***	I (1)	−2.206	−20.740***	I (1)	2002M10; 2008M08.
EUR	−1.676	−11.091***	I (1)	−3.304	−9.891***	I (1)	2002M04; 2008M04; 2013M07
JPY	−1.943	−19.284***	I (1)	−1.023	−15.734***	I (1)	1982M12; 1989M07; 1995M07; 2001M06; 2009M01
AG	−2.262	−16.433***	I (1)	−0.478	−15.677***	I (1)	2003M03; 2009M01
CR	−1.889	−21.599***	I (1)	−1.660	−16.202***	I (1)	1982M12; 1988M11; 1994M09; 2002M01
EN	−1.649	−21.455***	I (1)	−0.864	−16.046***	I (1)	1982M12; 1988M11; 1994M09; 2002M01
GO	−1.338	−17.924***	I (1)	−1.663	−17.057***	I (1)	–
ID	−3.133	−19.071***	I (1)	−0.548	−15.500***	I (1)	1988M07; 1994M10; 2008M04

Note: CR, GO, AG, EN, and ID, are respectively, crude oil price index, price of gold, agricultural price index, energy price index, and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. ADF test is the Augmented Dickey Fuller test; NL test is the Narayan and Liu test; All the variables are expressed in natural logs; FD denotes First Difference; ***, **, * indicate the rejection of the null hypothesis of a unit root at 1%, 5% and 10%, respectively; the test regression for all the unit root tests includes intercept and trend; I(d) implies the order of integration, where d is the number of differencing required for a series to become stationary; All the variables are in their log forms. The breaks are determined using the Bai-Perron (2003) test.

Table 4
Persistence and endogeneity test results.

	Persistence					Endogeneity				
	EUR	AUD	BRP	CAD	JPY	EUR	AUD	BRP	CAD	JPY
Full-Sample Period										
CR	0.972***	0.997***	0.997***	0.997***	0.997***	−0.073	−0.203**	−0.139**	−0.089*	−0.054
GO	0.995***	0.998***	0.998***	0.998***	0.998***	−0.181	−0.261*	−0.324***	−0.199*	−0.023
AG	0.994***	0.997***	0.997***	0.997***	0.997***	−0.375*	−0.458*	−0.529**	−0.439***	−0.297
EN	0.974***	0.997***	0.997***	0.997***	0.997***	−0.127	−0.270***	−0.183**	−0.117*	−0.066
ID	0.986***	0.996***	0.996***	0.996***	0.996***	−0.110	−0.010	−0.103	0.039	−0.127

Note: ***, **, * imply statistical significance of coefficients at 1%, 5%, and 10% levels, respectively. This in turn indicates the absence of persistence effect and endogeneity bias in the predictive model.

Table 5
In-sample predictability for the baseline (commodity-based) model.

Predicted	Predictor				
	AG	CR	EN	GO	ID
AUD	−0.671*** (0.025)	−0.237*** (0.016)	−0.255*** (0.018)	−0.156*** (0.012)	−0.583*** (0.022)
BRP	−0.765*** (0.044)	−0.254*** (0.021)	−0.269*** (0.024)	−0.348*** (0.022)	−0.572*** (0.024)
CAD	−0.476*** (0.022)	−0.227*** (0.017)	−0.225*** (0.018)	−0.228*** (0.014)	−0.360*** (0.013)
EUR	−0.784*** (0.037)	−0.061** (0.030)	−0.017 (0.032)	−0.683*** (0.036)	−0.320*** (0.032)
JPY	−0.066 (0.045)	0.152*** (0.018)	0.184*** (0.019)	0.091*** (0.021)	−0.218*** (0.030)

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively. Only the bias-adjusted heteroscedasticity consistent estimate, i.e. $\lambda_{adj}P_{t-1}$ is reported here while other parameter estimates are suppressed. This is the only estimate required to determine predictability (see Westerlund and Narayan, 2012, 2015).

Table 6
Forecast evaluation using the RMSE statistics.

Predicted		Predictor				
		AG	CR	EN	GO	ID
AUD	In-Sample	0.187	0.312	0.321	0.309	0.205
	Out-Sample (h = 6)	0.316	0.920	0.971	0.448	0.549
	Out-Sample (h = 12)	0.319	0.898	0.943	0.470	0.531
BRP	In-Sample	0.104	0.089	0.087	0.108	0.095
	Out-Sample (h = 6)	0.156	0.036	0.039	0.213	0.111
	Out-Sample (h = 12)	0.202	0.091	0.083	0.259	0.083
CAD	In-Sample	0.120	0.167	0.172	0.168	0.113
	Out-Sample (h = 6)	0.046	0.346	0.375	0.045	0.204
	Out-Sample (h = 12)	0.036	0.315	0.344	0.039	0.191
EUR	In-Sample	0.092	0.102	0.109	0.105	0.091
	Out-Sample (h = 6)	0.031	0.224	0.280	0.017	0.145
	Out-Sample (h = 12)	0.025	0.240	0.293	0.028	0.173
JPY	In-Sample	32.909	24.458	23.699	29.392	35.033
	Out-Sample (h = 6)	24.567	37.240	40.620	26.454	14.936
	Out-Sample (h = 12)	22.161	33.659	36.960	23.969	14.466

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. RMSE is the Root Mean Square Error. HA is the historical average.

Table 7
Forecast evaluation of the commodity-based model against the historical average.

Predicted		Predictor				
		AG	CR	EN	GO	ID
AUD	In-Sample	0.105*** (0.008)	0.094*** (0.009)	0.091*** (0.009)	0.062*** (0.007)	0.104*** (0.008)
	Out-Sample (h = 6)	0.103*** (0.017)	0.249*** (0.044)	0.261*** (0.045)	0.135*** (0.021)	0.161*** (0.027)
	Out-Sample (h = 12)	0.084*** (0.010)	0.202*** (0.026)	0.211*** (0.027)	0.114*** (0.013)	0.127*** (0.017)
BRP	In-Sample	0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.002*** (0.001)	0.008*** (0.001)
	Out-Sample (h = 6)	−0.011*** (0.002)	0.018*** (0.003)	0.022*** (0.003)	−0.022*** (0.001)	0.040*** (0.002)
	Out-Sample (h = 12)	−0.016*** (0.002)	0.021*** (0.002)	0.025*** (0.002)	−0.032*** (0.003)	0.051*** (0.004)
CAD	In-Sample	0.029*** (0.003)	0.025*** (0.002)	0.022*** (0.002)	0.020*** (0.002)	0.033*** (0.002)
	Out-Sample (h = 6)	0.024** (0.007)	0.084** (0.023)	0.088** (0.022)	0.022** (0.007)	0.057** (0.016)
	Out-Sample (h = 12)	0.021*** (0.004)	0.072*** (0.012)	0.076*** (0.011)	0.020*** (0.004)	0.049*** (0.008)
EUR	In-Sample	0.035*** (0.004)	0.021*** (0.002)	0.021*** (0.002)	0.036*** (0.003)	0.033*** (0.002)
	Out-Sample (h = 6)	0.013*** (0.003)	−0.021*** (0.001)	−0.028*** (0.002)	0.009** (0.003)	−0.010*** (0.001)
	Out-Sample (h = 12)	0.013*** (0.002)	−0.024*** (0.002)	−0.032*** (0.003)	0.009*** (0.002)	−0.014*** (0.002)
JPY	In-Sample	5110.401*** (395.684)	5282.948*** (384.644)	5342.650*** (383.377)	5070.113*** (371.898)	5023.907*** (398.642)
	Out-Sample (h = 6)	2087.960*** (198.189)	2639.928*** (227.269)	2797.258*** (247.498)	2152.725*** (179.200)	1644.238*** (152.047)
	Out-Sample (h = 12)	2376.260*** (160.593)	2940.091*** (182.935)	3113.081*** (195.940)	2458.614*** (160.170)	1964.169*** (139.676)

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. A positive and statistically significant coefficient for the CW test implies that the larger model outperforms the smaller model, otherwise, it does not. The former has more parameters than the latter. The statistics reported are the regression coefficients obtained from the CW test equation while the values in parentheses () are the corresponding standard errors. ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively.

(Frankel, 2010).

However, the predictive power of commodity prices in forecasting exchange rate accurately is in doubt (Mese and Rogoff, 1983). Chen et al. (2010) and Ferraro et al. (2015) lend support to the findings of Mese and Rogoff (1983) particularly for the out-of-sample forecasts and note that exchange rate follows a random walk and cannot be predicted by changes in commodity prices. In fact, Ferraro et al. (2015) further note that the predictability, if at all it exists, can only become evident when daily frequency is used and therefore may diminish as lower data frequencies are utilized. Thus, forecasting exchange rate with commodity prices that will produce better forecast results than time series models has been a daunting task in the literature. Attempts to upturn the results have led to the introduction of the lagged term of the dependent variable as a regressor in the predictive model for exchange rate (see Moosa and Burns, 2014a,b,c for a review of some of the related papers). In our study however, we offer an alternative approach which allows for the salient features of the relevant variables in the predictive model and we hypothesize that such considerations will produce desirable results for forecasting exchange rate at least with commodity prices.

3. Methodology

Following Chen et al. (2010) and Ferraro et al. (2015), we begin our methodology by specifying a bivariate single predictive model where

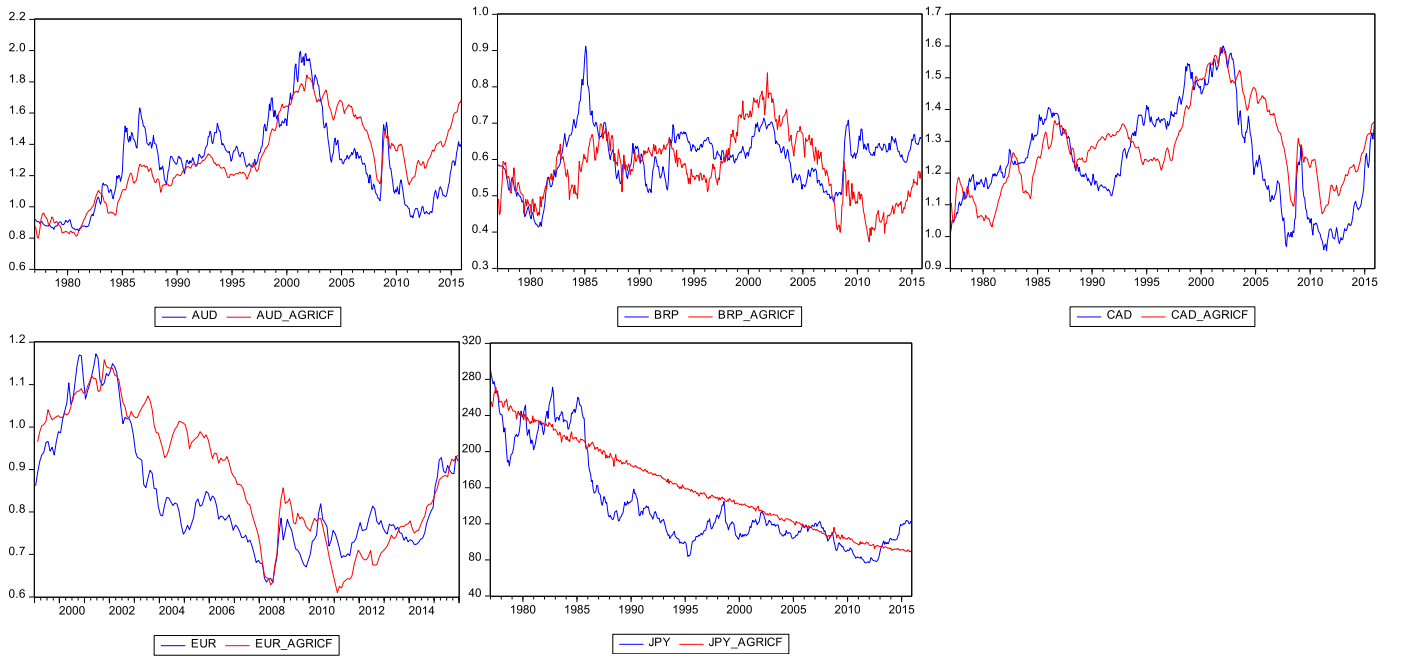


Fig. 6. Predictability graphs for predictive model – Baseline results [Agricultural Price Index].

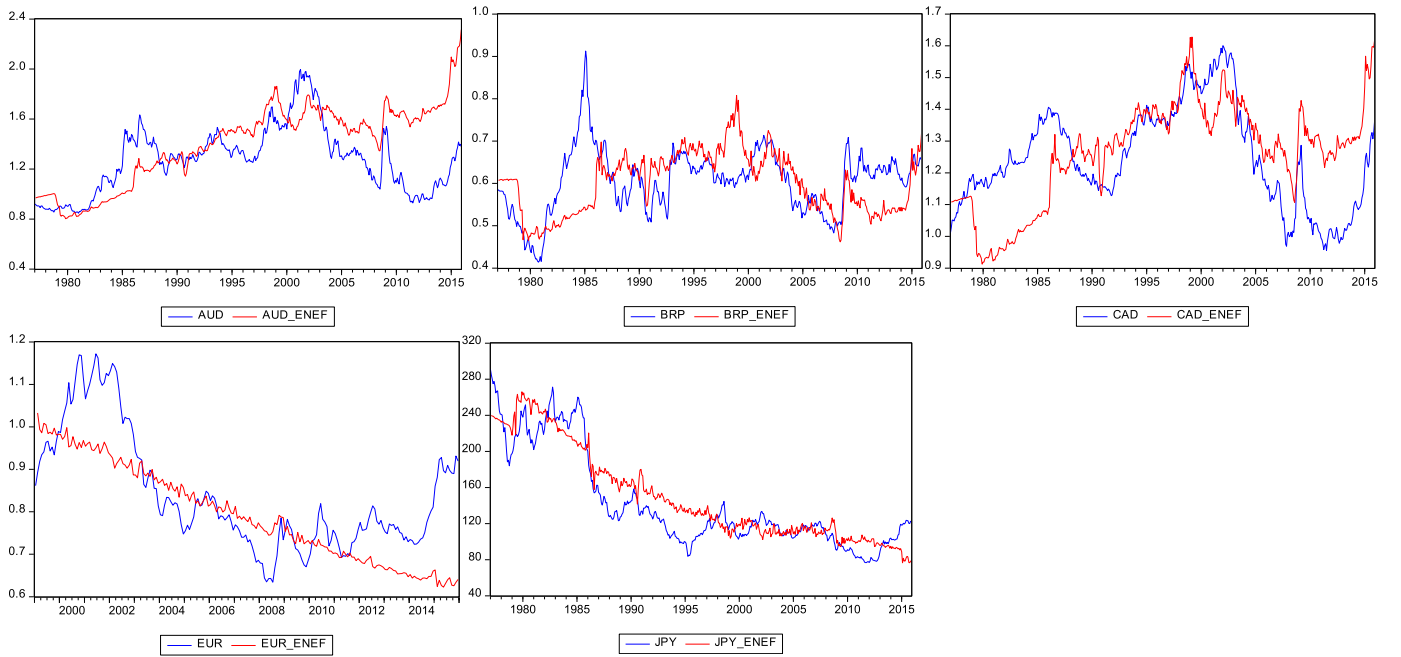


Fig. 7. Predictability graphs for predictive model – Baseline results [Energy Price Index].

commodity price is hypothesized as a predictor of exchange rate:

$$s_t = \alpha + \lambda p_{t-1} + \varepsilon_t \quad (1)$$

where s_t is the log of exchange rate using five most tradable currency pairs in the world involving the currencies of Australia, Canada, Japan, the United Kingdom and the Euro Area with US Dollar being the reference currency for the others; and p_t is the log of commodity prices where disaggregated components are used singly in the estimation process. Thus, for each currency pair, we have as many regressions as the number of commodity price indices considered (details about the data utilized are provided in the section that follows). The ε_t is zero mean idiosyncratic error term on exchange rate and the coefficient λ measures the relative impact of commodity prices on exchange rate and the underlying null hypothesis of no predictability is that $\lambda = 0$.

In order to resolve any probable endogeneity bias resulting from the correlation between p_t and ε_t as well as any potential persistence effect, we follow the approach of Lewellen (2004) and Westerlund and Narayan (2012, 2015). The underlying predictive model that accounts for these effects can be specified as:

$$s_t = \alpha + \lambda_{adj} p_{t-1} + \gamma (p_t - \rho_0 p_{t-1}) + \eta_t \quad (2)$$

where the parameter $\lambda_{adj} = \lambda - \gamma(\rho - \rho_0)$ is the bias adjusted OLS estimator of Lewellen (2004) which corrects for any persistence effect in the predictive model. The additional term $\gamma(p_t - \rho_0 p_{t-1})$ corrects for any endogeneity bias resulting from the correlation between p_t and η_t . Accounting for endogeneity bias here is important since there could be several determinants of exchange rate which are suppressed in equation (1). Such omissions could introduce endogeneity bias resulting from

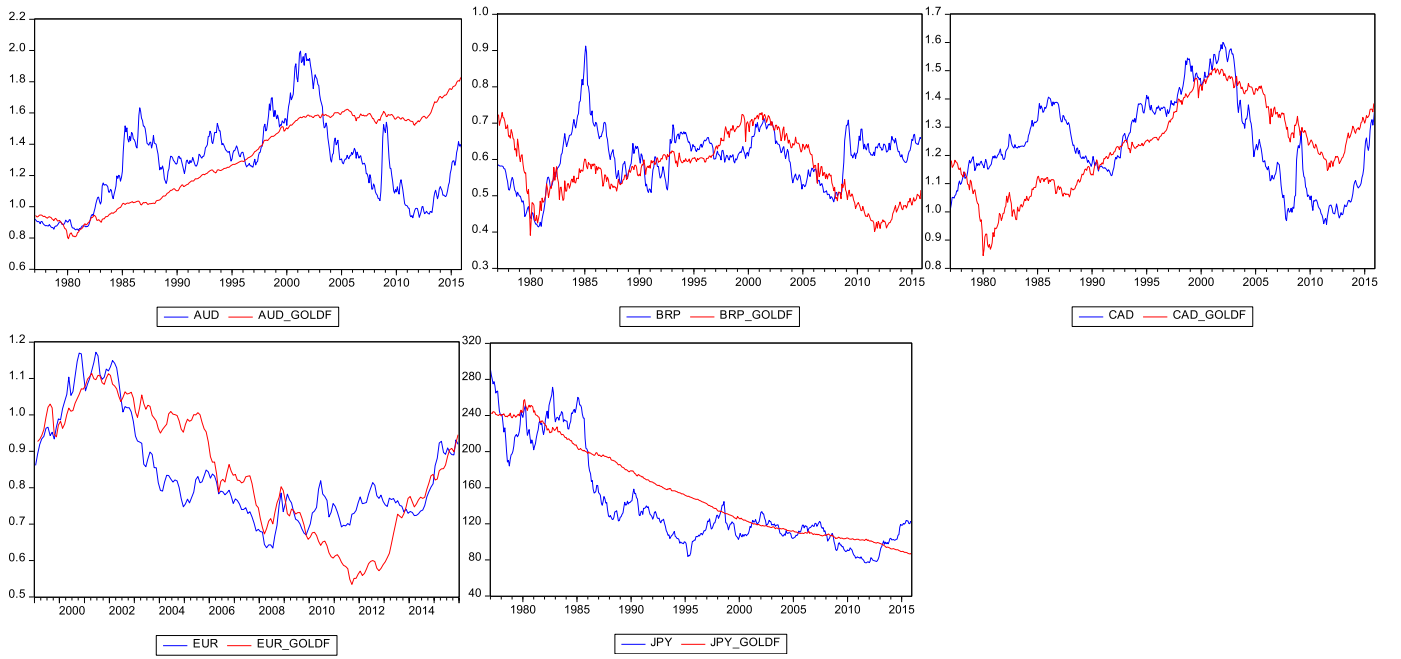


Fig. 8. Predictability graphs for predictive model – Baseline results [Gold Price Index]

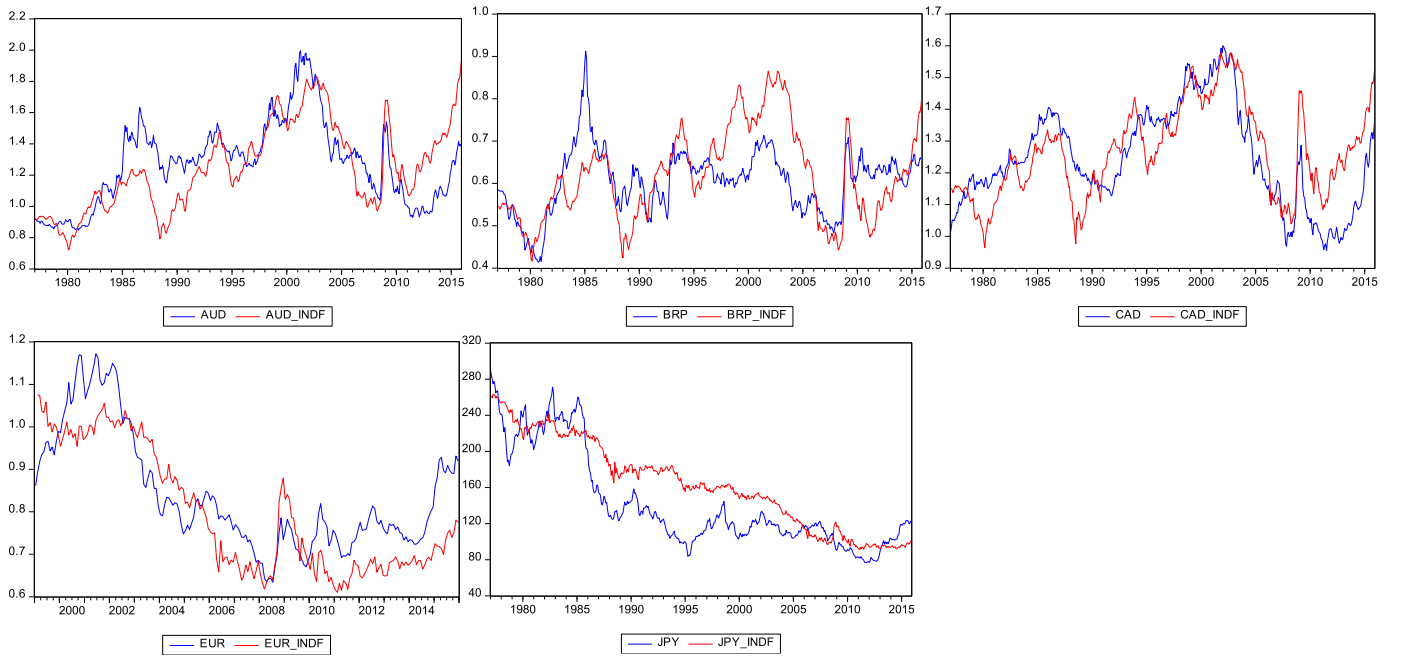


Fig. 9. Predictability graphs for predictive model – Baseline results [Industrial Material Price Index].

probable correlations between p_t and η_t . To resolve the conditional heteroscedasticity effect, WN (2012, 2015) suggest pre-weighting all the data by $1/\hat{\sigma}_{\eta}$ and estimating the resulting equation with the Ordinary Least Squares (OLS). This modified OLS estimator is described as the Feasible Quasi GLS estimator in WN (2012, 2015) and it is technically computed as:

$$\lambda_{adj}^{FQGLS} = \frac{\sum_{t=q_m+2}^T \tau_t^2 p_{t-1}^d s_t^d}{\sqrt{\sum_{t=q_m+2}^T \tau_t^2 (p_{t-1}^d)^2}} \quad (3)$$

where $\tau_t = 1/\sigma_{\eta,t}$ is used in weighting all the data in equation (2) and

$$p_t^d = p_t - \sum_{s=2}^T p_s / T.^4$$

In order to accommodate asymmetries, we decompose p_t into positive (p_t^+) and negative (p_t^-) changes using the Shin et al. (2014) approach,⁵ and thereafter we replace the p_t factor in equation (2) with the p_t^+ and p_t^- . Technically, the p_t^+ and p_t^- are respectively decomposed as:

⁴ See Westerlund and Narayan (2012, 2015) for computational details.

⁵ An alternative approach is the one proposed by Kilian and Vigfusson (2011), however, the Shin et al. (2014) is favoured because of its computational simplicity as well as its suitability for the analysis of long run and short run asymmetric effects in economic relationships.

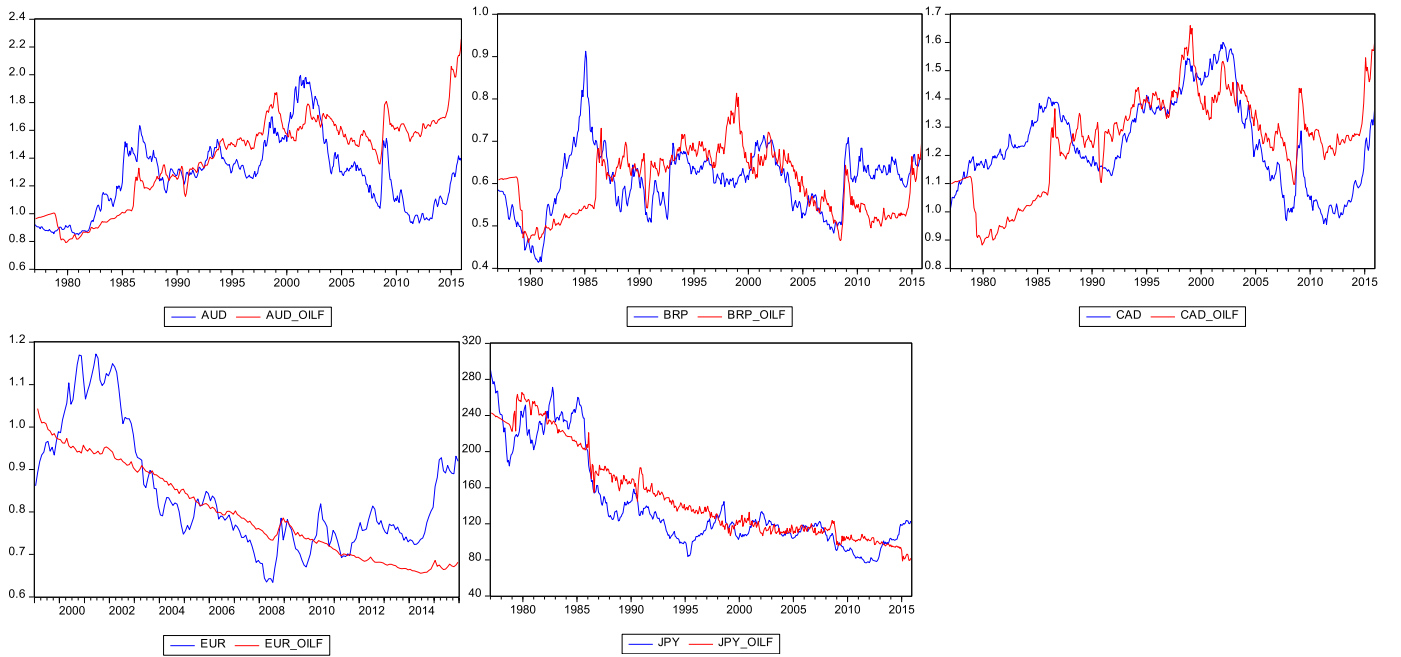


Fig. 10. Predictability graphs for predictive model – Baseline results [Crude Oil Price Index].

$$p_t^+ = \sum_{j=1}^t \Delta p_j^+ = \sum_{j=1}^t \max(\Delta p_j, 0) \quad (4)$$

$$p_t^- = \sum_{j=1}^t \Delta p_j^- = \sum_{j=1}^t \min(\Delta p_j, 0) \quad (5)$$

Therefore, p^+ and p^- can be defined as the positive and negative partial sum decompositions of commodity price changes, respectively. The revised predictive models for exchange rate that incorporate these asymmetric changes in commodity prices are given as:

$$s_t = \alpha + \lambda_{adj}^+ p_{t-1}^+ + \gamma^+ (p_t^+ - \rho_0 p_{t-1}^+) + \lambda_{adj}^- p_{t-1}^- + \gamma^- (p_t^- - \rho_0 p_{t-1}^-) + \eta_t \quad (6)$$

where equation (6) captures the role of positive and negative changes in commodity prices in the predictability of exchange rate. The null hypotheses of no predictability for p_t^+ and p_t^- respectively imply that $\lambda_{adj}^+ = 0$ and $\lambda_{adj}^- = 0$. Rather than attribute the shifts in exchange rate to asymmetric changes in commodity prices, we allow for significant structural breaks in the predictive model following the Bai-Perron test of structural shifts in linear regression models. Essentially, the procedure is to first determine any significant structural shift in equation (2) and thereafter we create a dummy variable for each break as an additional regressor in equation (2). Since Bai-Perron test allows for up to five breaks, we are able to identify more than one significant structural break for each proxy of commodity price which permits robustness checks with alternative break dates.⁶ Since exchange rate is the predicted series, the two optimal break points obtained for the latter are used in the estimation process. Thus, equation (2) is extended to include a dummy variable for structural break:

$$s_t = \alpha + \lambda_{adj} p_{t-1} + \gamma (p_t - \rho_0 p_{t-1}) + \phi D_{it} + \eta_t \quad (7)$$

where the inclusion of structural shift in the model is denoted by $D_t = 1$ if $t \geq \text{Break Date}$ and zero otherwise for all $i = 1, 2$. To verify whether the inclusion of the break is valid, we compare the forecast performance of equation (7) with equation (2) both for the in-sample and out-of-sample forecasts.

3.1. Estimation procedure and interpretation issues

The estimation procedure begins with the baseline model in equation (2) by estimating the impact of commodity prices on exchange rate using the FQGLS estimator as in (3). Thereafter, the extended variants in equations (6) and (7) that reflect the role of asymmetries and structural breaks are estimated and their forecasts (for both in-sample and out-of-sample periods) are comparatively evaluated. As customary in financial time series forecasting, we also assess the relative forecast performance of the historical average model which is nonfactor-based for completeness. The historical average involves regressing exchange rate (or its returns depending on the order of integration) on a constant. This is usually the benchmark model for most forecast analyses involving financial series (see also Westerlund and Narayan, 2012, 2015).

For the purpose of interpreting the impact of commodity prices on exchange rate behaviour, our focus is on the predictor series with the bias-adjusted parameter [for instance, $\lambda_{adj} p_{t-1}$ in equation (7)]. Other terms such as $\gamma (p_t - \rho_0 p_{t-1})$ and ϕD_t are only included to accommodate any inherent effect that may bias the regression estimates. Thus, in the predictability results, only the bias-adjusted parameter is reported for brevity while other estimates are suppressed. The extended model is estimated with the FQGLS estimator of WN (2015). Technically, this requires estimating the transformed model that allows for conditional heteroscedasticity and endogeneity with the Ordinary Least Squares (OLS) as previously described. It is important to note that the OLS employed here is different from the traditional variant as it already accommodates some additional features which are assumed to be non-existent in the traditional OLS estimator. If the parameter of interest, that is, λ_{adj} in the case of equation (7) is statistically significant, it then implies that the variable in question is a good predictor of the predicted series. However, this interpretation is only limited to the impact analysis of the predictor series. The estimated model is further subjected to both in-sample and out-of-sample forecast evaluation. This is where the role of accounting for asymmetries and structural breaks in the forecast accuracy of commodity-based exchange rate model is examined. In the next sub-section, we discuss the procedure followed for the forecast analyses in this study.

⁶ The computational advantages of using Bai-Perron (2003) structural break test are rendered in Narayan and Liu (2015) paper.

Table 8
In-sample predictability the commodity-based model with asymmetries.

Predicted	CR				EN				GO				ID			
	N	P	N	P	N	P	N	P	N	P	N	P	N	P	N	P
AG																
P																
AUD	−0.579*** (0.034)	−0.891*** (0.061)	−0.492*** (0.020)	−0.172*** (0.030)	−0.341*** (0.024)	−0.092** (0.042)	−0.263*** (0.015)	−0.535*** (0.016)	−0.383*** (0.029)	−0.126*** (0.043)						
BRP	−0.301*** (0.070)	−0.843*** (0.071)	−0.310*** (0.015)	0.185*** (0.020)	−0.293*** (0.017)	0.298*** (0.027)	−0.153*** (0.019)	−0.518*** (0.027)	−0.769*** (0.035)	−0.459*** (0.058)						
CAD	−0.273*** (0.031)	−0.532*** (0.035)	−0.299*** (0.015)	−0.247*** (0.021)	−0.275*** (0.019)	−0.168*** (0.023)	−0.004 (0.010)	−0.085*** (0.015)	−0.351*** (0.018)	−0.228*** (0.020)						
EUR	−0.504*** (0.052)	−1.957*** (0.113)	−0.005 (0.047)	−0.007 (0.043)	−0.238*** (0.042)	0.136** (0.057)	−0.021 (0.038)	−0.835*** (0.057)	−0.272*** (0.050)	−0.246*** (0.034)						
JPY	0.651*** (0.085)	−1.735*** (0.136)	0.136*** (0.026)	−0.058* (0.033)	0.116*** (0.026)	−0.089** (0.035)	0.552*** (0.043)	0.383*** (0.059)	0.038 (0.063)	−0.805*** (0.079)						

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. ***, **, and * indicate statistical significance at 1%, 5% and 10%, respectively. Only the bias-adjusted heteroscedasticity consistent estimate, i.e. $\lambda_{adj}P_{-1}$ is reported here while other parameter estimates are suppressed. This is the only estimate required to determine predictability (see Westerlund and Narayan, 2012, 2015).

3.2. Forecast procedure

As customary, the forecast evaluation is carried out for both the in-sample and out-of-sample periods. As previously mentioned, the need to conduct a distinct analysis for out-of-sample forecast is usually motivated by the conclusion of Bossaerts and Hillion (1999), Goyal and Welch (2003), Brennan and Xia (2005), Butler et al. (2005), and Ang and Bekaert (2007) indicating that return predictability is predominantly in-sample; and it is not robust to out-of-sample evaluations. For convenience, the pseudo forecasts approach is employed where the available (full sample) data (T) is partitioned into in-sample (say 's') and out-of-sample (say 'h') periods where h is the period ahead of in-sample (see also Narayan and Gupta, 2015; Bannigidadmath and Narayan, 2015; Narayan and Bannigidadmath, 2015; Devpura et al., 2018; Salisu et al., 2018a, 2019). However, there is no formal rule for partitioning data into in-sample and out-of-sample periods. Nonetheless, researchers have considered 25%, 50% and 75% (Narayan and Gupta, 2015). However, in order to ensure that the in-sample analyses cover the global financial crisis episode for the purpose of structural break test, we consider the period 1977:M01 to 2015:M12 for the estimation and the balance therefrom, i.e., 2016:M01 to 2017:M12, are used for the out-of-sample forecasts.⁷ For the out-of-sample forecast analyses, we utilize multiple periods namely 6 months, 12 months and 18 months for robustness. The start period for the in-sample estimation is carefully chosen in order to avoid including the period of fixed exchange rate regime globally which preceded the chosen start period. This is also evident in the graphs presented in Figs. 1–5 where all the selected exchange rates are observed to be fixed over the period preceding 1977:M01.

Two forecast measures are used to evaluate the in-sample and out-of-sample forecasts: the Root Mean Square Error (RMSE) and the Clark and West (2007) test for equality of forecasts for two nested models. The RMSE can be illustratively computed as $\sqrt{1/N \sum_{t=1}^T (\hat{r}_t - r_t)^2}$ where N is number of predictions used in computing the mean. Since the analyses involve comparing nested models, the Clark and West (2007) [CW hereafter] is more appropriate. This is because equation (2) and the extended models in equations (6) and (7) as well as the historical average model are nested. For instance, by restricting the parameters on the dummy variables to zero, equation (7) reduces to equation (2). The same intuition can be extended to equations (2) and (6) including the historical average model. However, the same cannot be said for equations (6) and (7). The statistic obtained from the CW test can be used to establish any significant difference in the forecast of any two nested models. The underlying procedure for the CW test involves calculating equation (8):

$$\hat{f}_{s+h} = (r_{s+h} - \hat{r}_{1,s+h})^2 - [(r_{s+h} - \hat{r}_{2,s+h})^2 - (\hat{r}_{1,s+h} - \hat{r}_{2,s+h})^2] \quad (8)$$

Where h is the forecast period, $(r_{s+h} - \hat{r}_{1,s+h})^2$ is the squared error for the restricted model (i.e., model 1), $(r_{s+h} - \hat{r}_{2,s+h})^2$ is the squared error for the unrestricted model (model 2), and $(\hat{r}_{1,s+h} - \hat{r}_{2,s+h})^2$ is the adjusted squared error introduced by the CW test to correct for any noise associated with the larger model's forecast. Thus, the sample average of $\hat{f}_{s+h}$ can be expressed as $MSE_1 - (MSE_2 - \text{adj.})$, and each term is computed as

$$MSE_1 = N^{-1} \sum (r_{s+h} - \hat{r}_{1,s+h})^2,$$

$$MSE_2 = N^{-1} \sum (r_{s+h} - \hat{r}_{2,s+h})^2, \text{ and}$$

$$\text{adj.} = N^{-1} \sum (\hat{r}_{1,s+h} - \hat{r}_{2,s+h})^2,$$

where N is the number of predictions used in computing these

⁷ In fact, the study of Ferraro et al. (2015) on the predictability of commodity prices in forecasting exchange rate finds that the predictive power of commodity prices is robust to the choice of the in-sample window size, and it does not depend on the sample period under consideration albeit for daily frequency.

Table 9
Forecast evaluation of the commodity-based model with asymmetries.

Predicted		Predictor				
		AG	CR	EN	GO	ID
AUD	In-Sample	0.003*** (0.001)	0.057*** (0.007)	0.070*** (0.006)	0.200*** (0.014)	0.023*** (0.003)
	Out-Sample (h = 6)	−0.029*** (0.006)	0.767*** (0.017)	1.216*** (0.091)	0.548*** (0.031)	0.514*** (0.023)
	Out-Sample (h = 12)	−0.033*** (0.003)	0.763*** (0.010)	1.116*** (0.058)	0.591*** (0.024)	0.478*** (0.022)
BRP	In-Sample	0.010*** (0.001)	0.008*** (0.001)	0.007*** (0.001)	0.027*** (0.003)	−0.005*** (0.000)
	Out-Sample (h = 6)	0.045*** (0.005)	−0.022** (0.006)	−0.017 (0.009)	0.043*** (0.002)	−0.000 (0.001)
	Out-Sample (h = 12)	0.052*** (0.003)	−0.042*** (0.007)	−0.045*** (0.010)	0.047*** (0.002)	−0.006** (0.002)
CAD	In-Sample	−0.001** (0.000)	0.004*** (0.001)	0.011*** (0.001)	0.008*** (0.000)	0.012*** (0.001)
	Out-Sample (h = 6)	0.004*** (0.000)	0.024* (0.010)	0.114*** (0.016)	−0.002* (0.001)	0.000 (0.002)
	Out-Sample (h = 12)	0.003*** (0.000)	0.016** (0.006)	0.086*** (0.012)	−0.003*** (0.001)	−0.000 (0.001)
EUR	In-Sample	0.022*** (0.002)	0.000*** (0.000)	0.012*** (0.002)	0.013*** (0.002)	0.003*** (0.000)
	Out-Sample (h = 6)	0.004** (0.001)	0.011*** (0.002)	0.109*** (0.005)	0.000 (0.000)	0.004** (0.001)
	Out-Sample (h = 12)	0.002** (0.001)	0.010*** (0.001)	0.106*** (0.003)	−0.000 (0.000)	0.007*** (0.002)
JPY	In-Sample	358.388*** (115.619)	133.937*** (45.737)	347.134*** (46.000)	605.898*** (102.350)	822.004*** (108.645)
	Out-Sample (h = 6)	1577.066*** (112.322)	1132.925*** (235.971)	449.075** (161.647)	−47.681** (13.296)	589.132*** (128.912)
	Out-Sample (h = 12)	1190.027*** (135.443)	970.844*** (161.059)	401.889*** (111.521)	−31.324* (16.577)	470.582*** (86.822)

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. A positive and statistically significant coefficient for the CW test implies that the larger model outperforms the smaller model, otherwise, it does not. The former has more parameters than the latter. The statistics reported are the regression coefficients obtained from the CW test equation while the values in parentheses () are the corresponding standard errors. ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively.

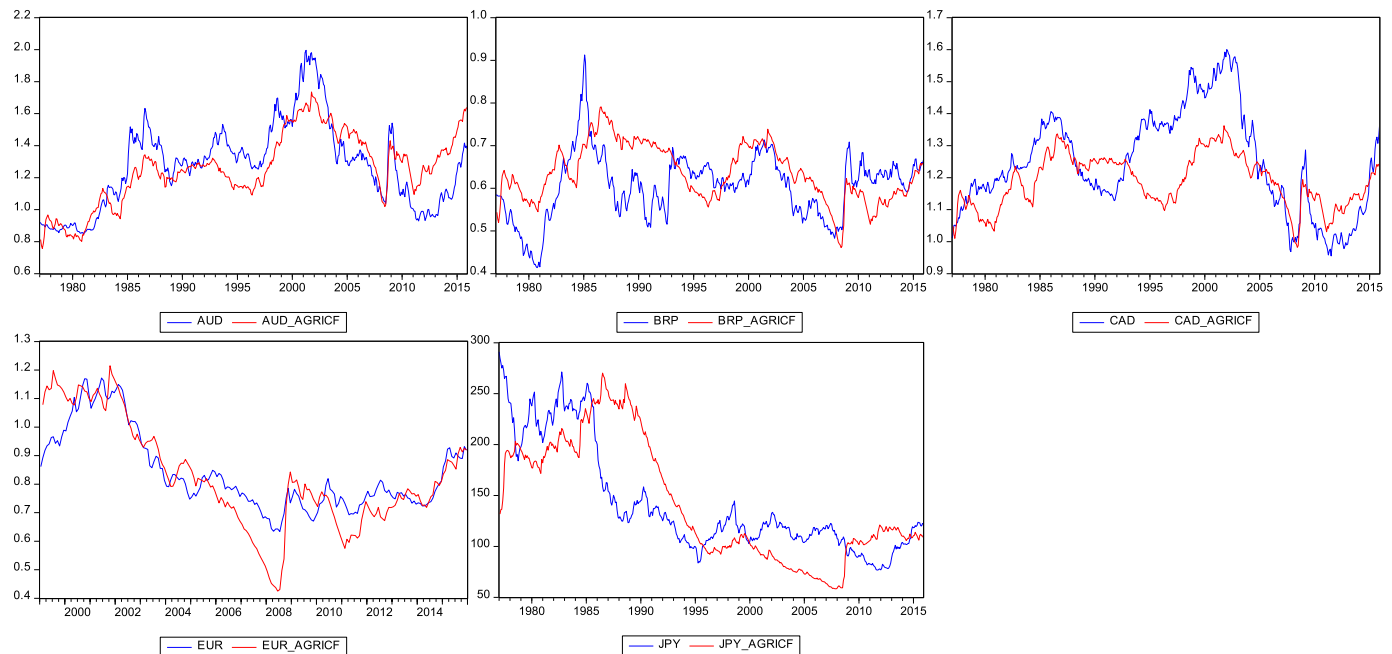


Fig. 11. Predictability graphs for predictive model with asymmetries [Agricultural Price Index].

averages. To test for equality of forecast performance between models 1 and 2 on the basis of the CW test, the $\hat{f}_{s+h}$ is regressed on a constant, and the resulting t-statistic for a zero coefficient is used to determine significance. Since the null hypothesis tests for equality of MSEs, the alternative hypothesis implies otherwise.

4. Data and preliminary analyses

4.1. Data source and description

The variables employed in this paper are commodity price indices and exchange rates (defined as domestic currency per unit of the US dollar). The sample set includes Australia, Canada, Japan, the United Kingdom, and the Euro Area. These countries and the Euro Area are members of the Organization for Economic Cooperation and

Development (OECD, subsequently) whose domestic currencies were mostly traded for the US dollar on the world foreign exchange market (Bank for International Settlement, BIS, 2016). As a result, the most traded currency pairs, according to Bank for International Settlement (BIS) (2016), include the exchange rates of the Euro, the Japanese Yen, the British Pound, the Australian and the Canadian dollar against the US dollar. This, therefore, necessitates the exclusion of the United States which serves as the reference country. Monthly data on commodity prices and exchange rates were obtained from various sources. The data on commodity prices cover the period of January, 1977 to October, 2017, and were sourced from the World Bank Database. All commodity prices except the price of gold (denominated in the US\$) are commodity price indices (representing the weighted average of prices of various commodities in each of the four groups). The various commodity price indices considered are the agricultural price index (AG),

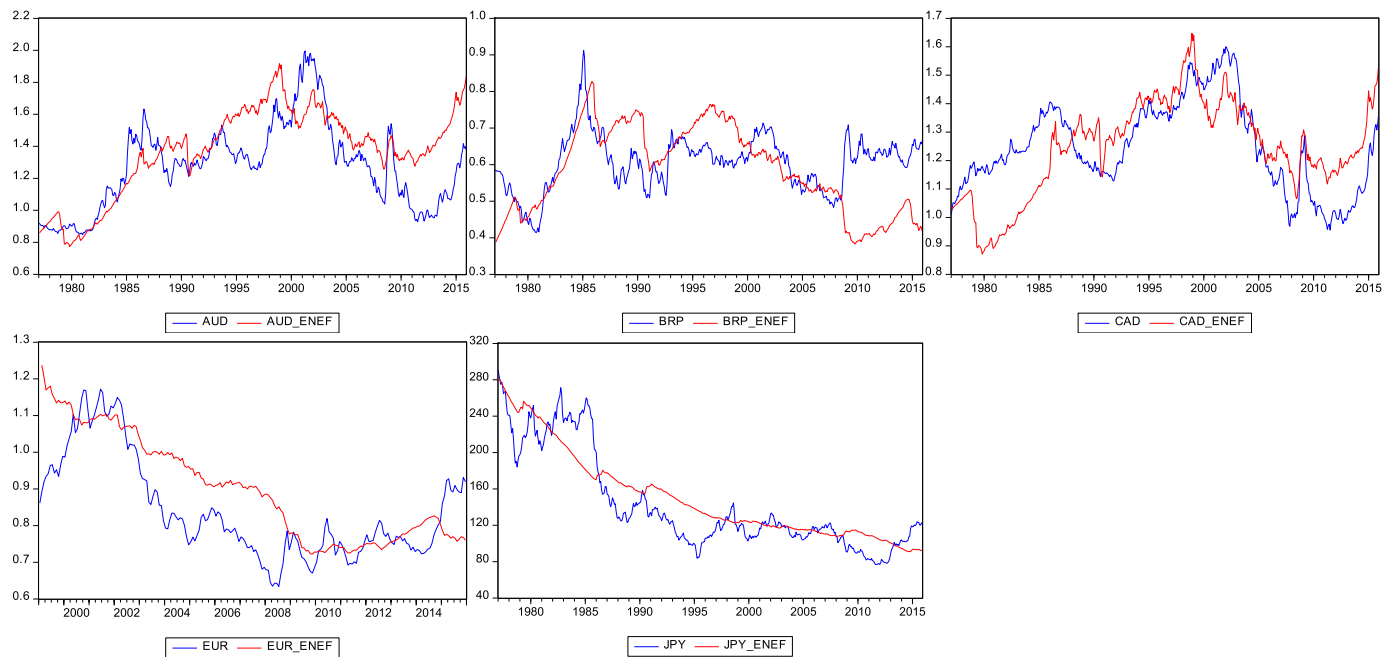


Fig. 12. Predictability graphs for predictive model with asymmetries [Energy Price Index].

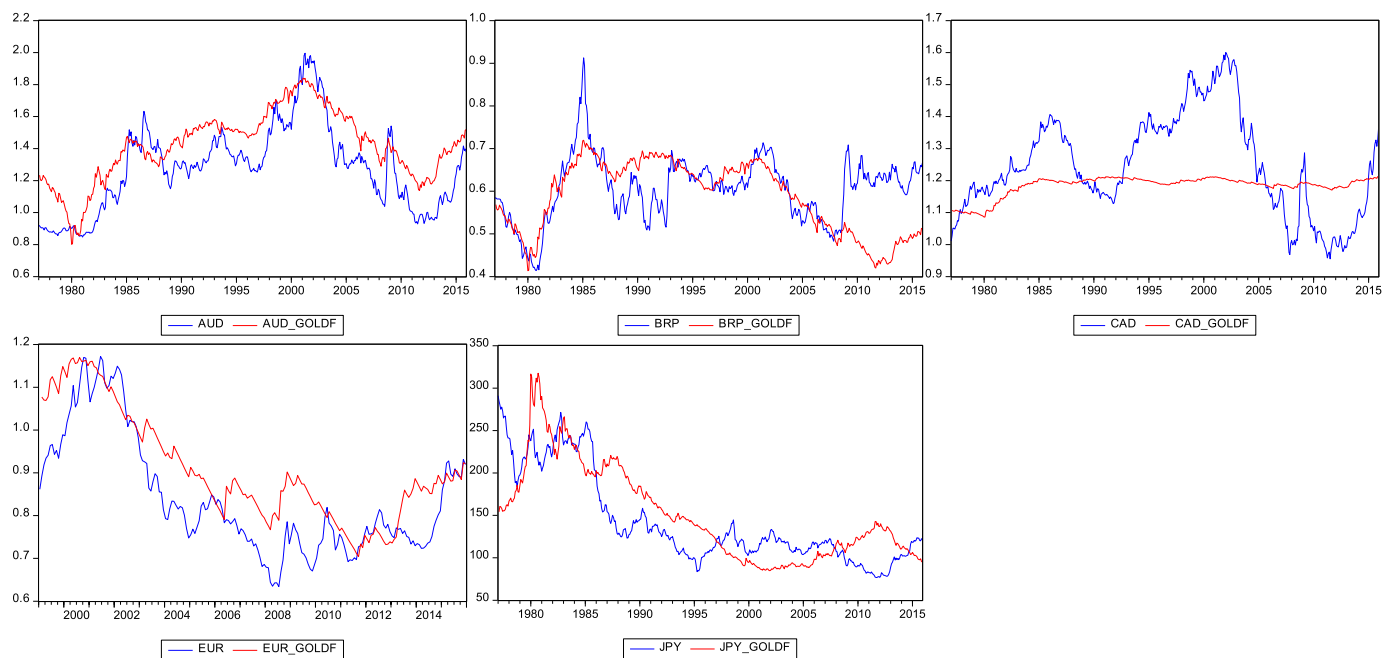


Fig. 13. Predictability graphs for predictive model with asymmetries [Gold Price Index]

crude oil price index (CR), energy price index (EN), price of gold (GO), and industrial metal price index (ID).

The data on exchange rates were obtained from the International Monetary Fund's International Financial Statistics (IMF-IFS, subsequently), and are also available for the period of January, 1977 to October, 2017, except for Euro/US dollar exchange rate that has a starting period of January, 1999. The shift in the world financial system from fixed exchange rate to flexible exchange rate in 1973 informs the choice of our start date (that is, January, 1977). Table 1 presents the data scope and the corresponding number of observations.

4.2. Preliminary analysis results

4.2.1. Graphical representation

Here, we explore the direction of co-movements between the five most traded currency pairs and each of the commodity price indices in separate graphs as contained in Figs. 1–5. As shown in Fig. 1, commodity prices and the Australian dollar/US dollar exchange rate moved in the same direction until 2002. Beyond this period, commodity prices and the Australian dollar/US dollar exchange rate moved in the opposite direction. This reflects the presence of asymmetric co-movements of both series over the sample period. Similar trend behaviour is noted in the case of the Canadian dollar/US dollar exchange rate and commodity

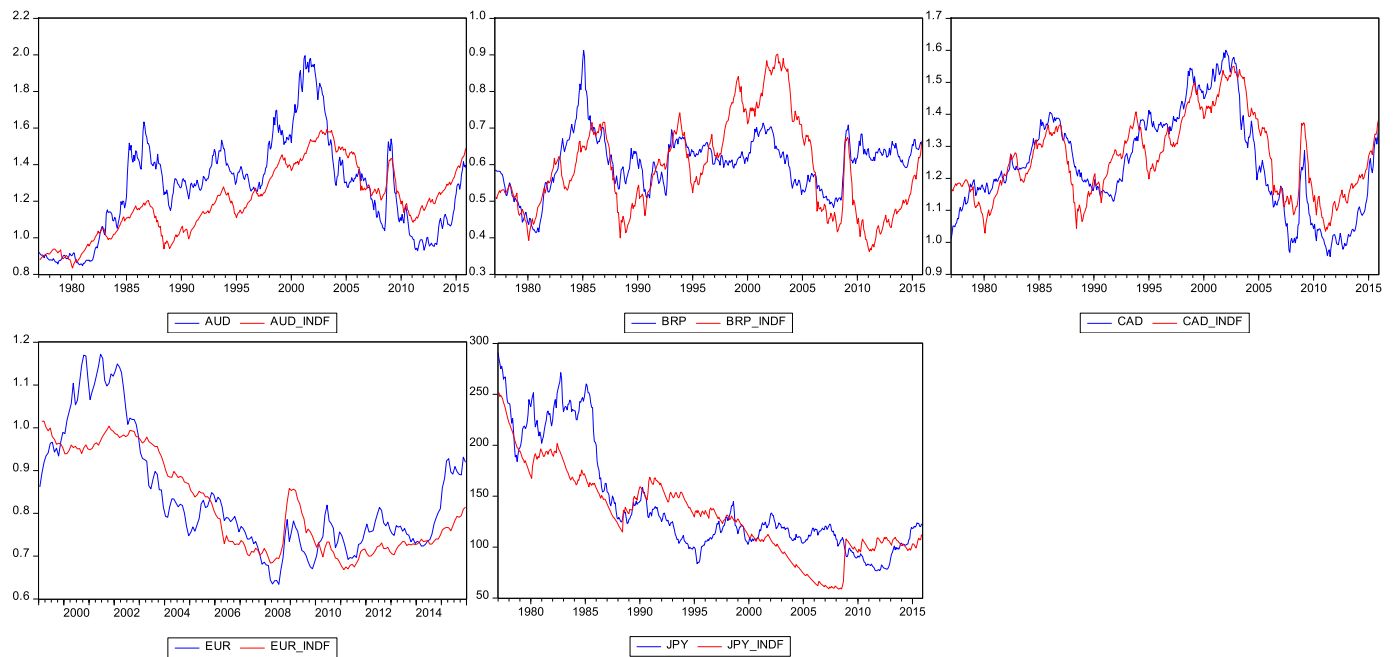


Fig. 14. Predictability graphs for predictive model with asymmetries [Industrial Material Price Index].

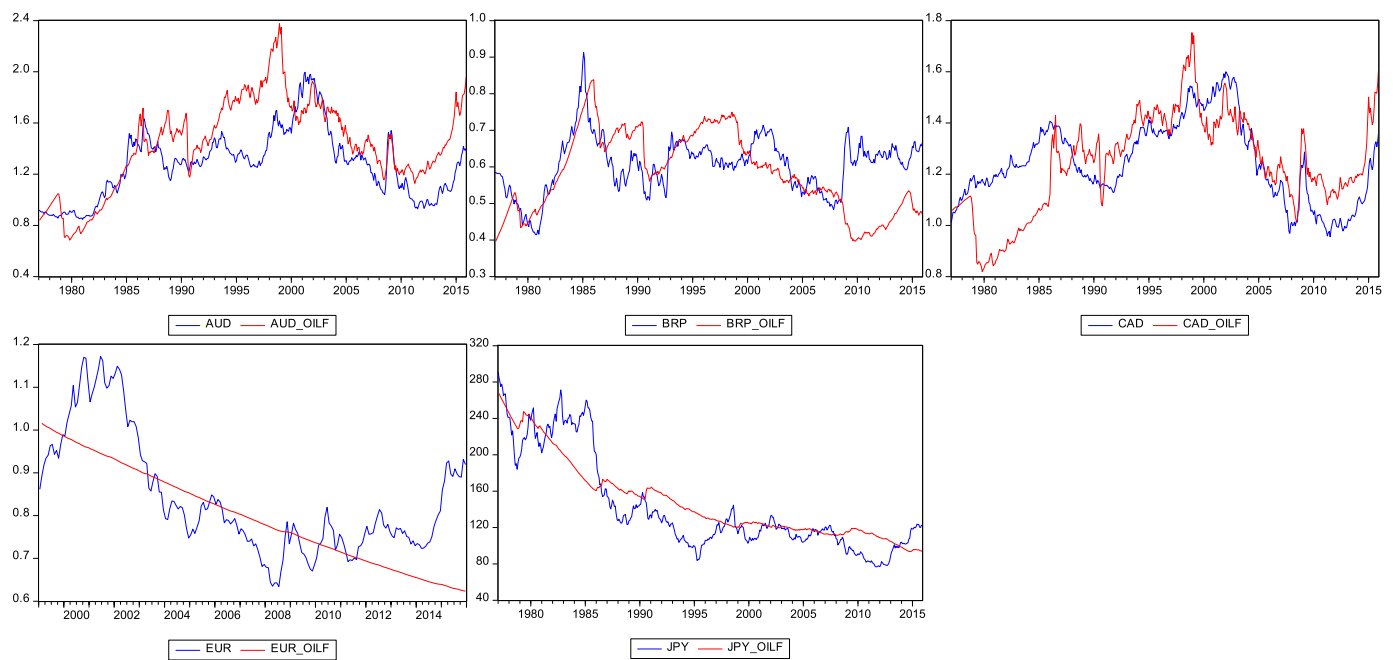


Fig. 15. Predictability graphs for predictive model with asymmetries [Crude Oil Price Index].

prices as shown in Fig. 3. Also, in Fig. 2, it is observed that commodity prices and the British pound/US dollar exchange rate had a positive co-movement. Commodity prices and British pound/US dollar exchange rate, however, exhibited dissimilar trend behaviors starting from 2002; as rising commodity prices were associated with the appreciation of the British pounds against the US dollar, and vice versa.

Similarly, in Fig. 4, we observe that commodity prices and the Euro-US dollar exchange rate co-moved in an inverse direction initially, but a positive co-movement between the two series was obtained later. All these imply that a rise in commodity prices was associated with an appreciation of the euro against the US dollar, and vice versa. In the same vein, commodity prices and the Japanese Yen/US dollar exchange rate exhibited an inverse co-movement (see Fig. 5); this implies that

increases in commodity prices were complemented by falling Yen against the US dollar (that is, appreciation of Japanese Yen), and vice versa. It therefore remains an empirical exercise to examine if accounting for asymmetry in commodity price movements and structural break matter in the predictability of exchange rate movements. This is the main thrust of the present paper.

Beyond the empirical exercise conducted in this study, it is important to highlight other underlying issues that may have characterized the observed trends in the individual currency pairs as evident in their respective graphs. This is essential because we do not want to be deluded by the co-movement between commodity prices and exchange rate to the extent that the behaviour in the latter can be wholly attributed to the former. This certainly will be an erroneous conclusion

Table 10
In-sample predictability for the commodity-based model with structural breaks.

Predicted	Predictor				
	AG	CR	EN	GO	ID
AUD	−0.347*** (0.025)	−0.091*** (0.009)	−0.097*** (0.010)	−0.087** (0.006)	−0.199*** (0.019)
BRP	−0.986*** (0.040)	−0.343*** (0.023)	−0.366*** (0.026)	−0.423*** (0.020)	−0.589*** (0.022)
CAD	−0.277*** (0.025)	−0.089*** (0.015)	−0.096*** (0.016)	−0.095*** (0.015)	−0.238*** (0.010)
EUR	−0.700*** (0.046)	0.019 (0.029)	0.077** (0.032)	−0.433*** (0.057)	−0.260*** (0.020)
JPY	−0.218*** (0.036)	−0.012 (0.016)	−0.012 (0.018)	−0.073*** (0.015)	−0.270*** (0.029)

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively. Only the bias-adjusted heteroscedasticity consistent estimate, i.e. $\lambda_{adj}P_{t-1}$ is reported here while other parameter estimates are suppressed. This is the only estimate required to determine predictability (see Westerlund and Narayan, 2012, 2015).

Table 11
Forecast evaluation of the commodity-based model with structural breaks.

Predicted		Predictor				
		AG	CR	EN	GO	ID
AUD	In-Sample	0.037*** (0.002)	0.154*** (0.014)	0.167*** (0.015)	0.125*** (0.008)	0.062*** (0.005)
	Out-Sample (h = 6)	0.201*** (0.015)	1.803*** (0.105)	2.016*** (0.101)	0.399*** (0.012)	0.727*** (0.019)
	Out-Sample (h = 12)	0.194*** (0.008)	1.689*** (0.063)	1.876*** (0.068)	0.417*** (0.009)	0.661*** (0.032)
BRP	In-Sample	0.014*** (0.001)	0.004*** (0.001)	0.003*** (0.001)	0.015*** (0.001)	0.006*** (0.001)
	Out-Sample (h = 6)	0.102*** (0.007)	0.000 (0.010)	−0.011 (0.010)	0.134*** (0.003)	−0.032*** (0.002)
	Out-Sample (h = 12)	0.130*** (0.010)	0.036** (0.012)	0.026* (0.013)	0.163*** (0.009)	−0.014** (0.006)
CAD	In-Sample	0.006*** (0.001)	0.021*** (0.002)	0.023*** (0.002)	0.021*** (0.001)	0.011*** (0.001)
	Out-Sample (h = 6)	0.003 (0.002)	0.247*** (0.026)	0.277*** (0.028)	0.002 (0.002)	0.082*** (0.006)
	Out-Sample (h = 12)	0.002* (0.001)	0.208*** (0.018)	0.237*** (0.019)	0.003* (0.001)	0.074*** (0.005)
EUR	In-Sample	0.007*** (0.001)	0.014*** (0.002)	0.017*** (0.002)	0.012*** (0.001)	0.011*** (0.001)
	Out-Sample (h = 6)	−0.001 (0.000)	0.089*** (0.003)	0.126*** (0.002)	0.001 (0.000)	0.043*** (0.002)
	Out-Sample (h = 12)	−0.000 (0.000)	0.100*** (0.004)	0.137*** (0.005)	0.001** (0.000)	0.054*** (0.005)
JPY	In-Sample	1139.286*** (118.045)	420.339*** (79.337)	372.719*** (75.922)	806.763*** (99.434)	1236.012*** (108.705)
	Out-Sample (h = 6)	1062.245*** (67.622)	2201.966*** (190.698)	2615.973*** (185.676)	1349.222*** (117.705)	232.112*** (29.386)
	Out-Sample (h = 12)	923.923*** (72.114)	1888.145*** (156.194)	2263.412*** (165.106)	1193.119*** (97.979)	191.061*** (23.496)

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. A positive and statistically significant coefficient for the CW test implies that the larger model outperforms the smaller model, otherwise, it does not. The former has more parameters than the latter. The statistics reported are the regression coefficients obtained from the CW test equation while the values in parentheses () are the corresponding standard errors. ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively.

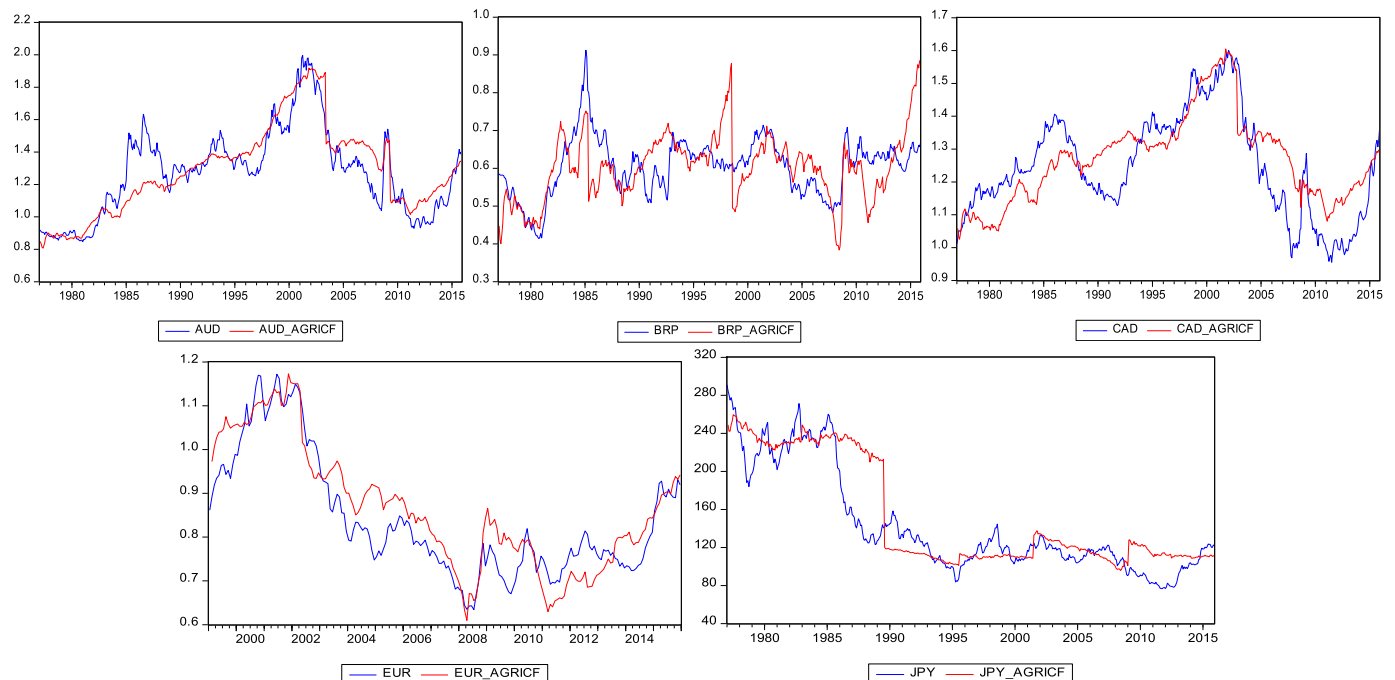


Fig. 16. Predictability graphs for predictive model with structural breaks [Agricultural Price Index].

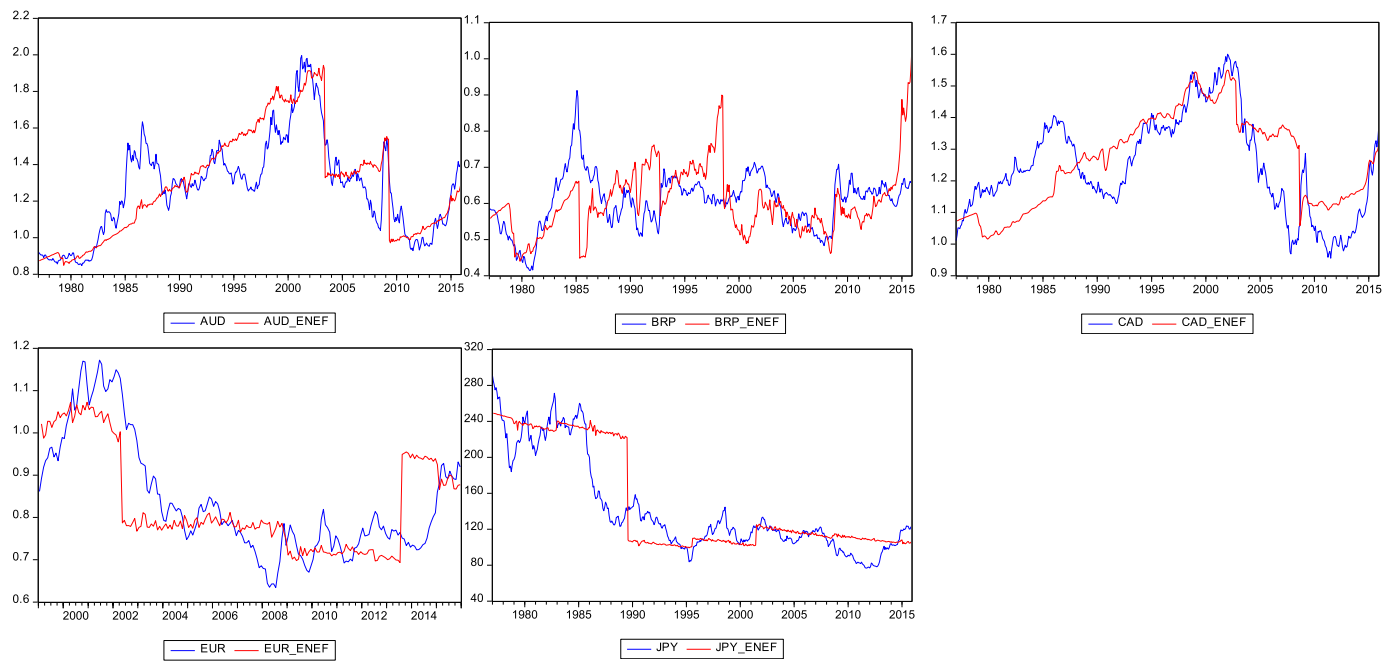


Fig. 17. Predictability graphs for predictive model with structural breaks [Energy Price Index].

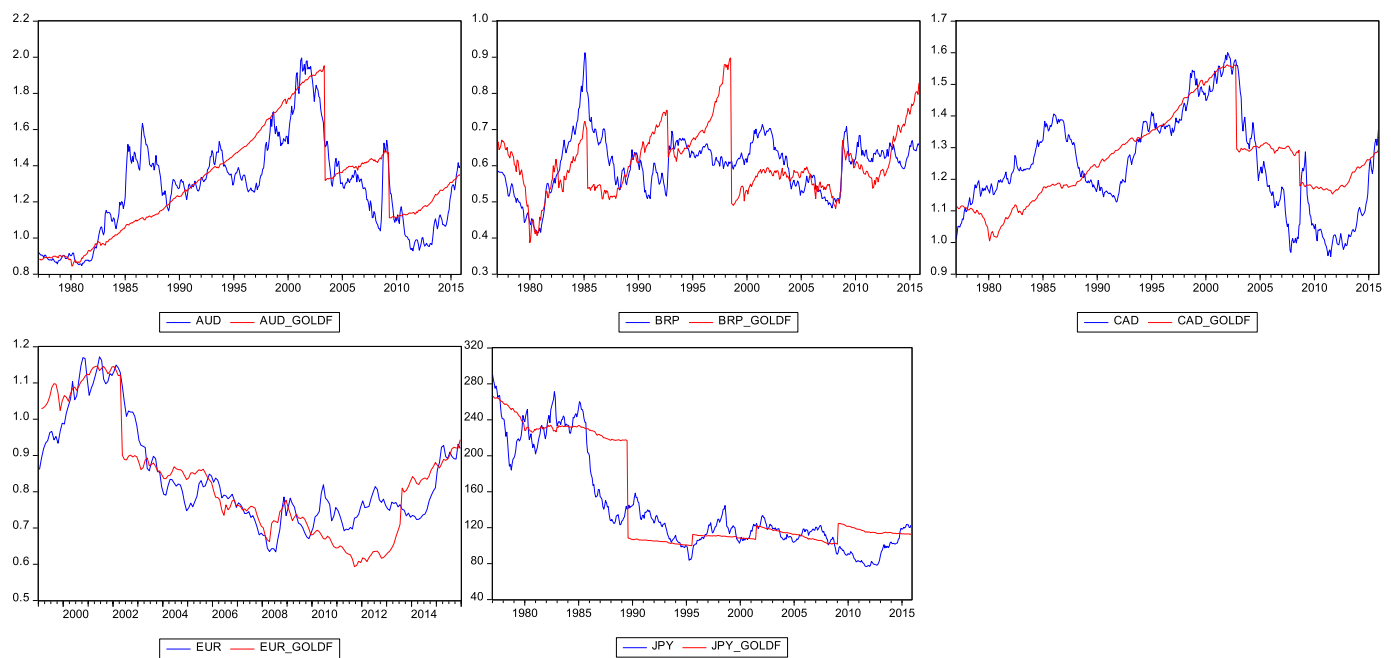


Fig. 18. Predictability graphs for predictive model with structural breaks [Gold Price Index].

and we are conscious of this fact. One common feature of all the currency pairs is that they are prominently described as floating exchange rates after 1976 when they shifted from the fixed regime and that explains why the earliest start period for our analyses is 1977. However, at different instances, they have witnessed internal and global shocks that affected their movements whether positively or negatively. Australian exchange rate for instance may have also been prominently influenced by interest rate among other macroeconomic fundamentals such as trade balance, economic growth and inflation. The exchange rate rose sharply in the late 1980s, 2000s and mid-2009, when Australian real interest rates were much higher than those overseas and the converse was noticed in the late 1990s when the real interest rates fell below those other major economies particularly the US. The British

Pound Sterling is another popular currency traded in the foreign exchange (forex) market. Like the Australian exchange rate, the UK currency also appears to respond to interest rate. The pound sterling slid to \$1.05 in February 1985 due to a higher interest rate offered by the US which strengthened the US dollar against the pound sterling. Subsequently, the pound rebounded as the UK economy boomed with low inflation and reasonable interest rates that offered a decent return for investors. This trend lasted for about a decade until the economic boom was replaced with the financial crisis of 2008 which saw the pound collapse as the UK economy fell into recession. The pound fell 35% to lows of about \$1.40 in early 2009, before picking up as the UK slowly emerged from the downturn (Broad, 2016). Like the previous currencies, the Canadian dollar has also witnessed ups and downs over

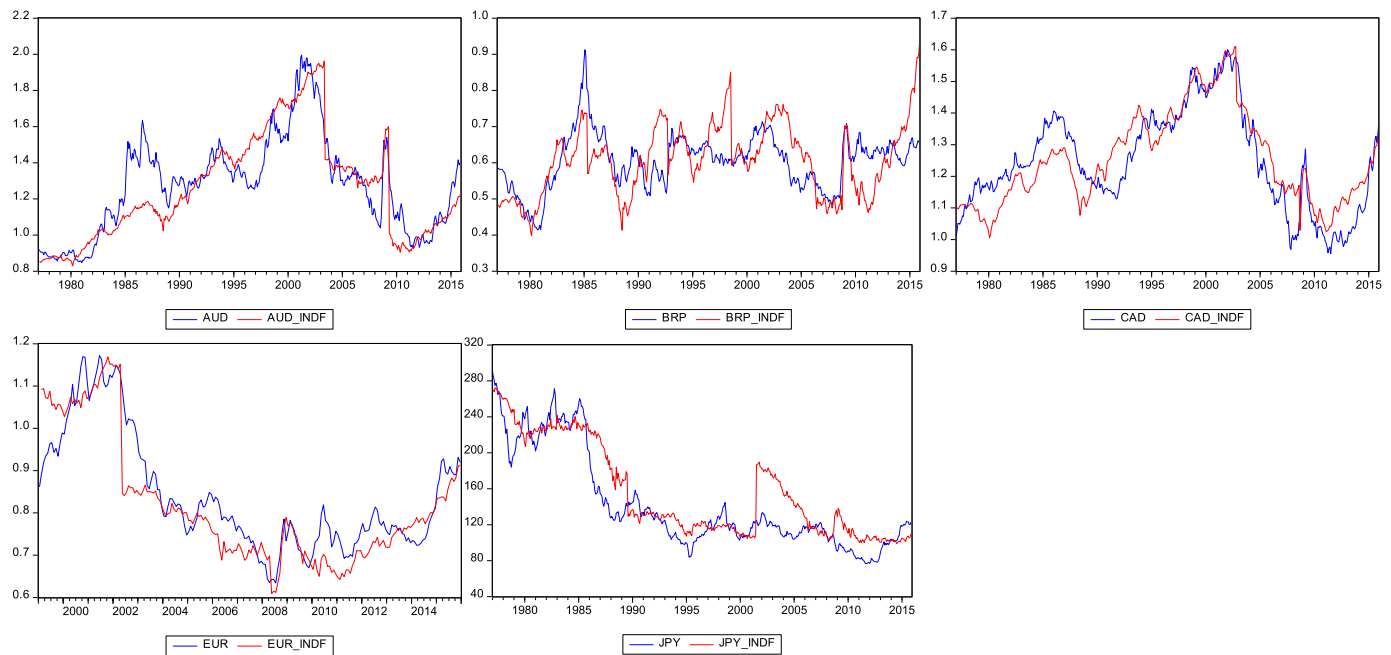


Fig. 19. Predictability graphs for predictive model with structural breaks [Industrial Material Price Index].

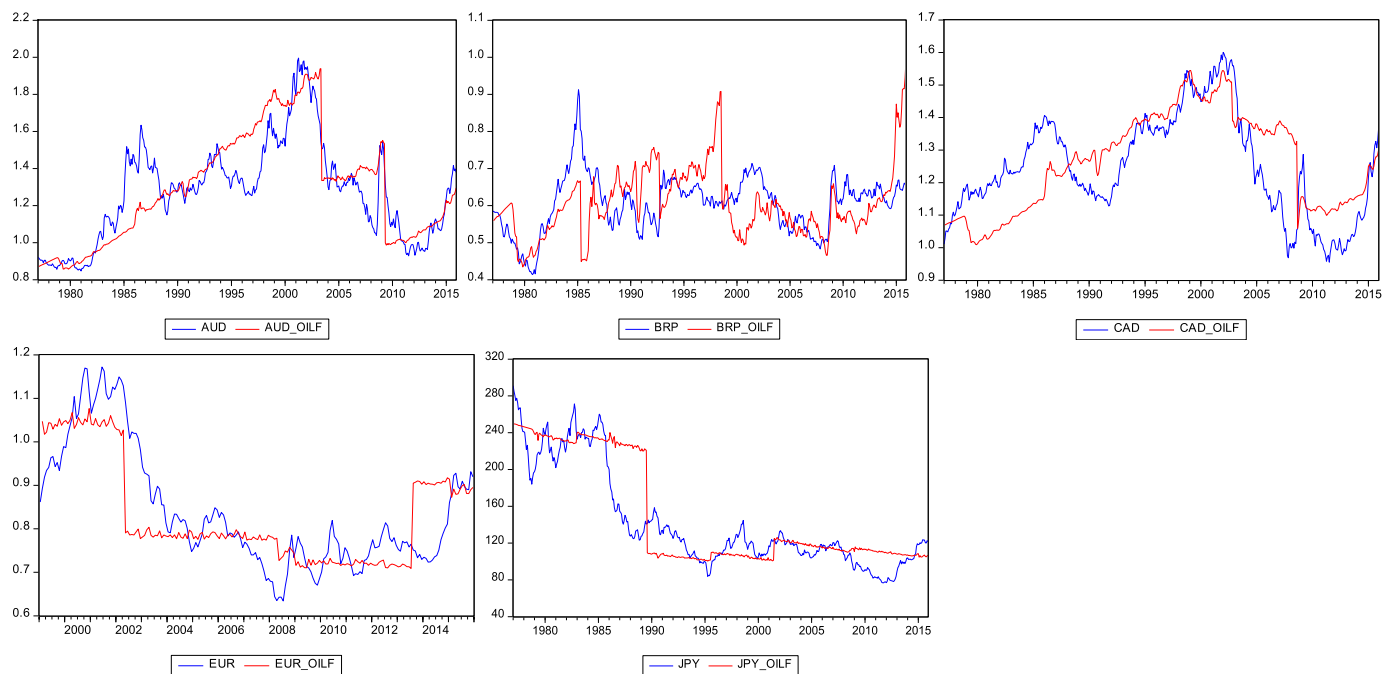


Fig. 20. Predictability graphs for predictive model with structural breaks [Crude Oil Price Index].

the study period and these trends also seem to respond to some internal and global shocks. The high rate of interest in Canada to about 21.2% in 1981 saw the Canadian dollar hit an all-time low of 69.13 cents (US) on February, 1986 but later improved through the latter part of the 1980s and early 1990s. However, weaker commodity prices among others saw a downward path for the Canadian dollar. It latter appreciated through 2003 to 2007, underscored by the price boom of Canada's commodity exports and in fact reached parity with the US dollar in September 2007 for the first time in about thirty years. Thus, the Canadian dollar seems to have been largely influenced by the swings in commodity prices.

The Euro/USD exchange rate is considered to be the most traded currency pair in the world given the currencies involved which represent the world's two largest economic and trading blocs and

therefore many multinational corporations prefer to conduct business in both the Europe and United States. Broadly speaking, the Euro/USD exchange rate responds to policy actions undertaken by both the European Central Bank and Fed. For instance, in September 2007 when Fed cuts Fed funds rate by 50 basis points, Euro strengthened against dollar. Similarly in December 2008 when Fed cuts rates to near zero, Euro strengthened against dollar. For the Japanese yen, its appreciation since the second half of the 1980s may also be attributed to sizable current account surpluses. All these highlighted potential factors that could influence exchange rate have been empirically evaluated in the literature (see for a survey of the literature, [Moosa and Burns, 2014a,b,c](#)). In any case, these factors are specific to the individual currency pairs and can therefore be described as “individual factors”.

Table 12

The combined model vs. the model with asymmetries only.

Predicted		Predictor				
		AG	CR	EN	GO	ID
AUD	In-Sample	0.030*** (0.002)	0.114*** (0.009)	0.049*** (0.004)	0.021*** (0.002)	0.043*** (0.002)
	Out-Sample (h = 6)	0.119*** (0.009)	0.330*** (0.073)	0.275*** (0.022)	0.023*** (0.003)	0.039*** (0.004)
	Out-Sample (h = 12)	0.123*** (0.005)	0.260*** (0.042)	0.254*** (0.014)	0.025*** (0.002)	0.040*** (0.003)
BRP	In-Sample	0.006*** (0.000)	0.012*** (0.001)	0.015*** (0.001)	0.012*** (0.001)	0.012*** (0.001)
	Out-Sample (h = 6)	0.017*** (0.002)	0.104*** (0.001)	0.153*** (0.002)	0.065*** (0.001)	0.020*** (0.003)
	Out-Sample (h = 12)	0.035*** (0.006)	0.122*** (0.006)	0.169*** (0.006)	0.078*** (0.004)	0.046*** (0.009)
CAD	In-Sample	0.002** (0.001)	0.007*** (0.001)	0.004*** (0.001)	0.013*** (0.001)	0.005*** (0.000)
	Out-Sample (h = 6)	0.012** (0.003)	0.092*** (0.010)	0.029*** (0.002)	0.008** (0.002)	0.004* (0.002)
	Out-Sample (h = 12)	0.010*** (0.002)	0.076*** (0.007)	0.026*** (0.002)	0.007*** (0.001)	0.002* (0.001)
EUR	In-Sample	0.009*** (0.001)	0.010*** (0.002)	0.008*** (0.002)	0.006*** (0.001)	0.002*** (0.001)
	Out-Sample (h = 6)	0.000 (0.001)	0.115*** (0.004)	−0.024*** (0.004)	−0.002* (0.001)	0.028*** (0.001)
	Out-Sample (h = 12)	0.002** (0.001)	0.119*** (0.003)	−0.023*** (0.002)	−0.004*** (0.001)	0.032*** (0.002)
JPY	In-Sample	2676.257*** (189.429)	771.186*** (88.803)	313.374*** (63.377)	2996.982*** (215.072)	2321.557*** (136.140)
	Out-Sample (h = 6)	71.216** (23.501)	115.541* (46.728)	944.528*** (134.542)	352.819** (132.355)	60.977 (30.415)
	Out-Sample (h = 12)	50.850 (35.983)	58.948* (28.680)	769.153*** (94.087)	274.495** (93.317)	80.060 (49.073)

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. A positive and statistically significant coefficient for the CW test implies that the larger model outperforms the smaller model, otherwise, it does not. The former has more parameters than the latter. The statistics reported are the regression coefficients obtained from the CW test equation while the values in parentheses () are the corresponding standard errors. ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively.

Table 13

The combined model vs. the model with structural breaks only.

Predicted		Predictor				
		AG	CR	EN	GO	ID
AUD	In-Sample	0.017*** (0.002)	0.026*** (0.003)	0.012*** (0.001)	0.080*** (0.011)	0.021*** (0.001)
	Out-Sample (h = 6)	0.000 (0.001)	0.007 (0.004)	0.003 (0.004)	0.026*** (0.003)	0.017** (0.005)
	Out-Sample (h = 12)	−0.000 (0.000)	0.005** (0.002)	0.002 (0.002)	0.032*** (0.004)	0.017*** (0.003)
BRP	In-Sample	0.006*** (0.000)	0.025*** (0.002)	0.056*** (0.005)	0.021*** (0.002)	0.005*** (0.001)
	Out-Sample (h = 6)	0.036*** (0.005)	0.370*** (0.031)	0.508*** (0.029)	0.018*** (0.003)	0.193*** (0.011)
	Out-Sample (h = 12)	0.031*** (0.003)	0.307*** (0.025)	0.446*** (0.025)	0.009*** (0.003)	0.151*** (0.015)
CAD	In-Sample	0.003*** (0.000)	−0.004*** (0.001)	−0.003*** (0.001)	0.004*** (0.001)	0.008*** (0.001)
	Out-Sample (h = 6)	0.001* (0.000)	−0.005*** (0.001)	−0.006*** (0.001)	−0.002*** (0.001)	0.001*** (0.001)
	Out-Sample (h = 12)	0.001*** (0.0002)	−0.004*** (0.001)	−0.004*** (0.001)	−0.001*** (0.000)	0.002*** (0.000)
EUR	In-Sample	0.009*** (0.001)	0.009*** (0.002)	−0.003* (0.001)	0.001* (0.001)	0.001 (0.001)
	Out-Sample (h = 6)	−0.000 (0.000)	0.054*** (0.005)	−0.014*** (0.004)	−0.004*** (0.000)	−0.001*** (0.000)
	Out-Sample (h = 12)	0.000 (0.000)	0.055*** (0.004)	−0.015*** (0.002)	−0.005*** (0.001)	−0.001* (0.000)
JPY	In-Sample	270.068*** (35.295)	758.368*** (138.559)	−22.164 (20.894)	655.879*** (123.788)	−17.560** (7.632)
	Out-Sample (h = 6)	2.104 (1.809)	277.594*** (46.808)	0.187 (11.093)	5.907 (3.343)	16.274*** (1.563)
	Out-Sample (h = 12)	13.262** (4.698)	411.445*** (72.722)	15.903 (9.894)	18.303*** (5.565)	14.957*** (1.758)

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. A positive and statistically significant coefficient for the CW test implies that the larger model outperforms the smaller model, otherwise, it does not. The former has more parameters than the latter. The statistics reported are the regression coefficients obtained from the CW test equation while the values in parentheses () are the corresponding standard errors. ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively.

What has however remained uninvestigated in the literature is the role of common factors such as those that are substantially driven by global demand and supply shocks. In other words, can all the currency pairs be driven by a common factor such as commodity prices? This is the contribution of this study.

4.2.2. Descriptive statistics

Table 2 reports the summary statistics for the global commodity price indices and exchange rate of four selected OECD countries and the Euro Area over the full sample period. In terms of standard deviation, the most volatile of the commodity prices is the price of gold while the least volatile is agricultural price index. Similarly, in the case of currency pairs, the average monthly dollar exchange rate is higher for Japanese yen than for other currencies, and the currency pair (Japanese yen-US dollar exchange rate) also appears to be the most volatile.

In addition, with respect to the statistical features of the series, the skewness is purely positive (indicating higher values than the mean) for

all the commodity price indices and exchange rates (except for British pound-US dollar exchange rate). However, the distribution is mixed for both the commodity price indices and exchange rate. In the case of the kurtosis statistics, the results are mainly leptokurtic across the commodity price indices. With respect to the exchange rates, the kurtosis statistics is majorly platykurtic across the full with the exception of Australian, British and Euro area exchange rates. Therefore, the Jarque-Bera statistic that tests for normality using information from kurtosis and skewness, suggests non-normality for all the commodity price indices. Also, the Jarque-Bera test statistic for exchange rate indicates non-normality for all the exchange rates with the exception of British Pound-US dollar exchange rate in the full sample.

4.2.3. Autocorrelation and conditional heteroscedasticity test results

Here, we conduct autocorrelation and heteroscedasticity tests using Ljung-Box test Q-statistics and Autoregressive conditional heteroscedasticity lagrangian multiplier (ARCH-LM) test F-statistics,

Table 14
In-sample predictability for quarterly and annual frequencies.

Predicted	Predictor	CR			EN			GO			ID		
		Y	Q	Y	Y	Q	Y	Y	Q	Y	Y	Q	Y
AUD	AG	−0.746*** (0.041)	−0.823*** (0.074)	−0.265*** (0.029)	−0.305 (0.054)	−0.280*** (0.032)	−0.418*** (0.064)	−0.156*** (0.021)	−0.245*** (0.049)	−0.576*** (0.038)	−0.618*** (0.061)	−0.513*** (0.044)	−0.641*** (0.128)
BRP	AG	−0.669*** (0.081)	−0.640*** (0.232)	−0.220*** (0.036)	−0.222** (0.085)	−0.228*** (0.040)	−0.243** (0.097)	−0.352*** (0.039)	−0.338*** (0.095)	−0.513*** (0.044)	−0.641*** (0.128)	−0.357*** (0.024)	−0.138** (0.062)
CAD	AG	−0.429*** (0.031)	−0.189** (0.072)	−0.227*** (0.030)	0.006** (0.048)	−0.214*** (0.032)	0.025 (0.051)	−0.224*** (0.023)	0.010 (0.027)	−0.357*** (0.024)	−0.138** (0.062)	−0.364*** (0.063)	−0.516*** (0.085)
EUR	AG	−0.678*** (0.095)	−0.368*** (0.101)	−0.175*** (0.052)	−0.424*** (0.055)	−0.165*** (0.057)	−0.475*** (0.070)	−0.668*** (0.056)	−0.187*** (0.051)	−0.364*** (0.063)	−0.516*** (0.085)	−0.243*** (0.055)	−0.691*** (0.105)
JPY	AG	−0.027 (0.081)	−0.731*** (0.251)	0.152*** (0.029)	−0.183 (0.131)	0.181*** (0.032)	−0.285** (0.137)	0.034 (0.036)	−0.251* (0.126)	−0.243*** (0.055)	−0.691*** (0.105)	−0.243*** (0.055)	−0.691*** (0.105)

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. ***, **, and * indicate statistical significance at 1%, 5% and 10%, respectively. Only the bias-adjusted heteroscedasticity consistent estimate, i.e. $\lambda_{adj}P_{t-1}$ is reported here while other parameter estimates are suppressed. This is the only estimate required to determine predictability (see Westerlund and Narayan, 2012, 2015). Q and Y denote quarterly and annual frequencies respectively.

respectively (see Table 2 below). We consider three different lag lengths (k) of 4, 8, and 12 for robustness. Our results show the presence of significant serial dependence and ARCH effects at both lower and higher orders for the commodity prices. We also find the presence of significant serial dependence and ARCH effects at both lower and higher orders for exchange rates. Our results are, therefore, robust across the various lag lengths considered.

4.2.4. The unit root test result

Two unit root tests are conducted in this study. The first is the conventional ADF-type unit root test and the second is the GARCH-based unit root test of Narayan and Liu (2015) and Narayan et al. (2016b). One of the advantages of using the latter is that it is suitable for series with trending behaviour, structural breaks and conditional heteroscedasticity.⁸ These features are not unexpected for the series in question given the outcome of the preliminary analyses in Table 2 as well as the figures presented.

The result of Augmented Dickey-Fuller (ADF) unit root test contained in Table 3 shows that all the ten variables (comprising five exchange rates and five commodity price indices) considered in this paper are integrated of order one; implying that the variables are non-stationary in levels, but that they achieve stationarity after first difference. By implication, the null hypothesis of a unit root is not rejected for both commodity prices and exchange rates. The non-stationarity of the series is further confirmed by the trend-GARCH-based unit root test with structural breaks. In other words, regardless of the technique used, the conclusion remains the same. These results equally authenticate the presence of persistence in the series which is one of the preconditions for using the Westerlund and Narayan (2012, 2015) estimator.

4.2.5. Persistence and endogeneity test results

Premised on the fact that the rejection of the null hypothesis of a unit root for the predictors, which are commodity prices in our own case, is not a sufficient condition to assume the absence of persistence, we further test for persistence and endogeneity in the predictors (see Table 4 below) over the full sample period. The persistence test has the null hypothesis of no persistence effect in the predictors. The coefficient of the AR (1) process was estimated for each predictor using OLS estimator and the results were found to be close or equal to one which is often the features of series with higher order of integration, thus, suggesting that the predictors possess high level of persistency. With regards to the endogeneity test, it revealed that all commodity price indices are endogenous with the exception of industrial metal price index. This, therefore, motivates our choice of the estimator that addresses the problem of any potential endogeneity bias that may arise from persistence and/or endogeneity of the predictors (commodity prices).

5. Discussion of results

The analyses and discussion of results are done as follows. First, we begin the analyses by exploring the in-sample predictability of the commodity-based (single-factor) model expressed in equation (2) where each of the commodity-specific indices is singly captured in the predictive model. The in-sample predictability is based on the coefficient of the bias-adjusted heteroscedasticity consistent parameter in the predictive model. A factor is considered to be a good predictor in this instance if the relevant coefficient is statistically significant, otherwise, it is not. Consequently, we evaluate the forecast performance of the commodity-based predictive model with the historical average, which is usually a benchmark model for most financial series (see for example

⁸ Other prominent unit root tests with structural breaks such as Lumsdaine and Papell (1997), Lee and Strazicich (2003), Perron (2006) and Narayan and Popp (2010, 2013) fail to account for conditional heteroscedasticity in their respective test regressions.

Table 15
Forecast evaluation of the role of alternative data frequencies.

Predicted	Predictor															
	AG				EN				GO				ID			
	Q	Y	Q	Y	Q	Y	Q	Y	Q	Y	Q	Y	Q	Y		
AUD	In-Sample	0.113*** (0.015)	0.114***(0.034)	0.098*** (0.014)	0.105*** (0.030)	0.095*** (0.015)	0.106*** (0.027)	0.061*** (0.012)	0.077*** (0.027)	0.104*** (0.014)	0.11603*** (0.033)					
	Out-Sample	0.075*** (0.002)	0.037 (0.049)	0.182***(0.007)	0.088 (0.122)	0.192***(0.005)	0.076 (0.094)	0.105***(0.008)	0.046 (0.074)	0.138***(0.006)	0.073 (0.100)					
	Sample															
	(h = 6)															
BRP	In-Sample	0.072*** (0.008)	0.056 (0.034)	0.171*** (0.016)	0.111 (0.074)	0.179*** (0.016)	0.091 (0.057)	0.104*** (0.012)	0.068 (0.048)	0.130*** (0.012)	0.109 (0.068)					
	Out-Sample															
	Sample															
	(h = 12)															
CAD	In-Sample	0.004*** (0.001)	0.004*(0.002)	0.003*** (0.001)	0.004*(0.002)	0.003*** (0.001)	0.004*(0.002)	0.002*(0.001)	0.004 (0.003)	0.007*** (0.001)	0.008*** (0.002)					
	Out-Sample	−0.013 (0.006)	−0.002 (0.002)	0.011 (0.004)	0.008 (0.006)	0.015 (0.005)	0.011 (0.007)	−0.027** (0.002)	−0.009*(0.001)	0.032** (0.001)	0.013 (0.010)					
	Sample															
	(h = 6)															
EUR	In-Sample	−0.013** (0.003)	−0.009 (0.006)	0.013*** (0.002)	0.002 (0.006)	0.016*** (0.003)	0.005 (0.007)	−0.031*** (0.004)	−0.022 (0.013)	0.040*** (0.005)	0.029 (0.017)					
	Out-Sample															
	Sample															
	(h = 12)															
JPY	In-Sample	0.028*** (0.005)	0.017** (0.007)	0.025*** (0.004)	0.001 (0.004)	0.020*** (0.004)	−0.002 (0.005)	0.020*** (0.004)	0.001 (0.001)	0.032*** (0.004)	0.018*** (0.005)					
	Out-Sample	0.016 (0.005)	0.005 (0.009)	0.054 (0.015)	0.002 (0.015)	0.058 (0.014)	0.001 (0.021)	0.017 (0.006)	0.003 (0.008)	0.039 (0.012)	0.007 (0.005)					
	Sample															
	(h = 6)															
JPY	In-Sample	0.016*** (0.002)	0.005 (0.005)	0.055*** (0.007)	0.005 (0.010)	0.060*** (0.007)	0.006 (0.013)	0.018*** (0.003)	0.002 (0.005)	0.041*** (0.005)	0.004 (0.004)					
	Out-Sample															
	Sample															
	(h = 12)															
JPY	In-Sample	0.024*** (0.003)	0.018** (0.006)	0.030*** (0.004)	0.041*** (0.013)	0.032*** (0.004)	0.040*** (0.014)	0.038*** (0.006)	0.021** (0.008)	0.036** (0.005)	0.039** (0.014)					
	Out-Sample	0.002 (0.003)	−0.008 (0.010)	−0.019*** (0.000)	0.002 (0.001)	−0.023** (0.001)	0.004 (0.001)	0.005 (0.001)	−0.005 (0.006)	−0.009* (0.001)	0.001 (0.002)					
	Sample															
	(h = 6)															
JPY	In-Sample	0.004 (0.002)	−0.010 (0.006)	−0.025** (0.006)	−0.002 (0.004)	−0.030** (0.007)	0.003 (0.001)	0.010 (0.006)	−0.010 (0.007)	−0.015* (0.005)	−0.005 (0.005)					
	Out-Sample															
	Sample															
	(h = 12)															
JPY	In-Sample	4843.535*** (661.078)	2092.603 (1295.573)	5580.335*** (646.205)	1279.462 (1239.580)	5620.017*** (645.850)	1573.261 (1226.645)	4649.105*** (650.104)	1800.691 (1289.640)	4764.956*** (659.440)	3161.017** (1266.139)					
	Out-Sample	2341.271 (481.860)	−424.587* (65.011)	3580.402 (599.117)	−1102.020** (74.962)	3694.135 (641.574)	−879.800 (175.091)	1872.706* (281.351)	−699.735** (45.469)	1748.763* (271.188)	223.926 (219.332)					
	Sample															
	(h = 6)															
JPY	In-Sample	2277.186*** (308.063)	−610.613* (189.775)	3554.638*** (448.559)	−1419.311** (320.230)	3669.451*** (469.404)	−1166.589* (304.084)	1897.697*** (248.424)	−812.908** (116.178)	1837.207*** (217.670)	42.757 (221.038)					
	Out-Sample															

Table 16
Forecast evaluation of the role of other determinants.

Predicted		Predictor				
		AG	CR	EN	GO	ID
BRP	In-Sample	0.006*** (0.001)	0.006*** (0.001)	0.007*** (0.001)	0.006*** (0.001)	0.006*** (0.001)
	Out-Sample (h = 6)	−0.003*** (0.001)	0.008*** (0.001)	0.016*** (0.001)	−0.012*** (0.001)	0.013*** (0.001)
	Out-Sample (h = 12)	0.001 (0.002)	0.016*** (0.003)	0.027*** (0.004)	−0.012*** (0.001)	0.022*** (0.003)
CAD	In-Sample	0.026*** (0.002)	0.031*** (0.002)	0.031*** (0.002)	0.031*** (0.002)	0.024*** (0.002)
	Out-Sample (h = 6)	0.003 (0.002)	0.098** (0.030)	0.106** (0.032)	0.026** (0.010)	0.070** (0.020)
	Out-Sample (h = 12)	0.005*** (0.002)	0.087*** (0.015)	0.095*** (0.016)	0.030*** (0.006)	0.067*** (0.010)
EUR	In-Sample	0.030*** (0.004)	0.025*** (0.004)	0.025*** (0.004)	0.030*** (0.004)	0.037*** (0.004)
	Out-Sample (h = 6)	0.014*** (0.002)	−0.019*** (0.001)	−0.030*** (0.002)	0.009*** (0.002)	0.004*** (0.002)
	Out-Sample (h = 12)	0.015*** (0.002)	−0.021*** (0.002)	−0.035*** (0.003)	0.011*** (0.002)	0.002*** (0.001)

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. A positive and statistically significant coefficient for the CW test implies that the larger model outperforms the smaller model, otherwise, it does not. The former has more parameters than the latter. The statistics reported are the regression coefficients obtained from the CW test equation while the values in parentheses () are the corresponding standard errors. ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively.

Narayan et al., 2016a; Salisu et al., 2019). Both the RMSE statistics and Clark and West (2007) test are used for the forecast comparison and are conducted for both the in-sample and out-of-sample periods. Based on the presumption that the commodity-based model will outperform the benchmark model, we move to the next phase of our analyses which involves comparing the former model with the extended variants that control for structural breaks and asymmetries both separately and jointly. Recent studies on financial time series forecasting albeit for different series allow for asymmetries (see Ferraro et al., 2015; Narayan and Gupta, 2015; Valadkhani and Smyth, 2017; Salisu and Isah, 2018); structural breaks (Pesaran and Timmermann, 2007; Pesaran et al., 2013; Salisu et al., 2019) or both (Salisu et al., 2019). The third phase of the analysis involves robustness test and it is carried out in two folds: (i) we control for other important determinants of exchange rate and (ii) we use alternative data frequency. We now take in turn each of the highlighted phases in subsequent sections.

5.1. In-sample predictability and forecast evaluation of the baseline (commodity-based) model

The predictability results for each of the commodity-specific indices are presented in Table 5. Two things are noteworthy in the results. First, commodity price is a good predictor of exchange rate regardless of the choice of the proxy used for the former. This conclusion is drawn from the predictability coefficients reported in the table which are largely found to be statistically significant except for two coefficients highlighted in red colour. Second, there seems to be a uniform negative sign for virtually all the coefficients. The only few exceptions are the three coefficients highlighted in blue colour which are specific to Japan. Thus, barring the latter country, we may conclude that commodity price is a good hedge for exchange rate risk. In addition, it also validates the theoretical expectation which assumes that higher commodity prices will lead to appreciation of the exchange rates of net commodity exporters while the reverse will be the case for net commodity importers. Put differently, higher commodity prices may necessitate the demand for the major tradable currency pairs such as those considered in this study which will in turn lead to their appreciation. The results of the in-sample predictability of commodity prices in exchange rate are consistent with the findings of Chen et al. (2010) and Ferraro et al. (2015) which reveal that the relationship between oil price and exchange rate is statistically significant at any standard level of significance, and for all frequencies considered in their paper (i.e., daily, monthly and quarterly). In fact, Ferraro et al. (2015) also report the same sign for oil price-exchange rate predictability for Canada.

Next, we evaluate the forecast performance of the Baseline (Commodity-based) model relative to the benchmark model using the

RMSE and CW test (see Tables 6 and 7 for the two measures respectively). We find that the baseline model significantly outperforms the benchmark model for the in-sample analyses regardless of the commodity price index used. The in-sample predictability graphs are presented for the five selected commodity prices in Figs. 6–10. Contrary to the evidence of Chen et al. (2010) and Ferraro et al. (2015), we are able to establish some improvements in the out-of-sample forecast. In fact, the latter study was only able to beat the random walk model for the out-of-sample case when daily frequency was utilized. In our study, the superiority of the commodity-based model relative to the benchmark model is observed for three of the five exchange rates. In this case, all the selected commodities predict exchange rates more accurately than the benchmark model (see Table 7). The three favored exchange rates in this regard are the Australian Dollar exchange rate (AUD), Canadian Dollar exchange rate (CAD) and the Japanese Yen exchange rate while for the remaining two exchange rates; the superiority of the commodity-based model over the benchmark model is noticed in three of the five commodities. The two cases where the latter performs better than the former are captured in bold in Table 7. Thus, for the affected exchange rates, the choice of commodity-specific index has implications on the forecast outcome of the predictive model.

5.2. Does asymmetry matter for the predictability of exchange rates?

This question is premised on the findings of Ferraro et al. (2015) and Salisu et al. (2019). While the former finds that accounting for asymmetries cannot upturn the superiority of the random walk model over the commodity (oil price)-based exchange rate model, the reverse is reported in the latter for stock returns model where oil price is also considered as a predictor. The in-sample predictability estimates are obtained from the estimation of equation (6) and are reported in Table 8. We find evidence of asymmetries in virtually all the estimates and majority of them are significant at the 1 percent level. The few cases of insignificance are expressed in bold. We further assess the relative superiority of the asymmetric model over the baseline (symmetric) model. This is done using the CW test and the results are presented in Table 9. The asymmetric model seems to outperform the symmetric model for the in-sample scenario irrespective of the commodity considered. Similarly, for the out-of-sample case, the symmetric model only appears superior for the British Pound Sterling to US dollar (BRP) albeit with three commodities namely Crude oil (CR), Energy and Industrial indices. All the other estimates are in favour of the asymmetric model. The in-sample predictability graphs are also presented in Figs. 11–15 in order to show how the asymmetric model is able to track the actuals. The graphs visually depict better predictive powers for most of the commodity prices when compared with those presented in

Figs. 6–10 for the baseline model. The significant role of asymmetric effect of commodity prices in the predictive model of exchange rate as reported here appears to support the assertion made by [Chen et al. \(2010\)](#) which states that “... the asymmetry in forecastability can be the result of many factors, ranging from potential non-linearities to the relative depth of the exchange rate markets, which may contribute to the exchange rates being more closely approximated by a random walk than commodity prices.”

Our results also align with the evidence of [Salisu et al. \(2019\)](#) which report similar evidence for stock return predictability. The underlying issues reflected in our predictive model which also mimic the model by [Salisu et al. \(2019\)](#) may be responsible for the consistency in the results. We may adduce the same fact to the inconsistency between our results and those of [Ferraro et al. \(2015\)](#). Thus, the choice of specification has implications on the forecast outcomes.

5.3. Do structural breaks matter for the predictability of exchange rate?

The need to account for parameter instability is increasingly becoming inevitable given its effect on the forecast results (see for example, of [Paye and Timmermann, 2006](#); [Rapach and Wohar, 2006, 2008](#); [Pesaran and Timmermann, 2007](#); [Giacomini and Rossi, 2009](#); [Chen et al., 2010](#); [Pesaran et al., 2013](#); [Salisu et al., 2019](#)). Thus, we attempt to examine same for our proposed commodity-based model for exchange rate by estimating equation (7) and the predictability results are reported in [Table 10](#). The break dates reported in [Table 3](#) are included as additional variables in the baseline model using dummy variable. The results are substantially consistent with the baseline model with respect to the direction of relationship indicating that the estimates are robust to structural breaks. Like the asymmetric case, we also compare the forecast performance of the structural break-augmented predictive model with the baseline model (see [Table 11](#)). The outcome is similar to the asymmetric variant which suggests the need to account for structural breaks in the commodity-based predictive model for forecasting exchange rate. This finding is consistent with the [Chen et al. \(2010\)](#) results at least for the in-sample scenario where better forecasts are reported by the latter after allowing for structural breaks. However, for the out-of-sample scenario, the results of [Chen et al. \(2010\)](#) seem to align with the [Mese and Rogoff \(1983\)](#) puzzle indicating that none of the fundamentals, including commodity prices, consistently forecasts exchange rate movements better than a random walk.⁹ On the contrary, we find strong out-of-sample predictability when structural breaks are reflected in the estimation and forecasts. The predictability graphs (see [Figs. 16–20](#)) are also indicative of the high predictive powers inherent in accounting for structural breaks when forecasting exchange rate. As previously mentioned, the distinctions between our study and the two related papers may be attributed to the choice of methodology and it may not be out of place to argue that ignoring some relevant features of the underlying fundamentals may result in unimpressive forecast outcomes for the theory-based models.

5.4. Does the combined model outperform models with asymmetries and structural breaks?

Since both models with asymmetries and structural breaks are found to exhibit better forecast performance than the baseline, it is only necessary to see if there is any possibility to accommodate both features in a predictive model for exchange rate in order to enhance its forecast performance. The model that incorporates these features is technically described in this study as the combined model for easy reference. The estimated model can be expressed as:

$$s_t = \alpha + \lambda_{adj}^+ p_{t-1}^+ + \gamma^+ (p_t^+ - \rho_0 p_{t-1}^+) + \lambda_{adj}^- p_{t-1}^- + \gamma^- (p_t^- - \rho_0 p_{t-1}^-) + \phi D_{it} + \eta_t \quad (9)$$

All the variables are as previously defined. The forecast comparison between the combined model and model with asymmetries on one hand is presented in [Table 12](#), while the one involving the model with structural breaks immediately follows in [Table 13](#). The CW test results indicate substantially the superiority of the combined model over other reduced variants albeit with more significant results over the asymmetric model. This tends to suggest that ignoring the presence of significant asymmetric effects may have less severe impact on the forecast results than ignoring the presence of significant structural breaks.

5.5. Other robustness tests

5.5.1. Using quarterly and annual frequencies

The consideration of an alternative frequency is motivated by the study of [Ferraro et al. \(2015\)](#) which finds that the predictability of commodity prices when forecasting exchange rate is more ephemeral and appears only for some commodities and only in daily data. The study concludes that for both the out-of-sample and in-sample analyses, the frequency of the data is important to detect the predictive ability of commodity prices, as the out-of-sample predictive ability breaks down when considering monthly and quarterly data.¹⁰ The results of the in-sample predictability and forecast evaluation are respectively presented in [Tables 14 and 15](#). While the in-sample estimates are consistent based on the direction of relationship, however the predictability appears to diminish with lower frequency. This is in line with the evidence of [Ferraro et al. \(2015\)](#) as well as other studies testing the role of data frequency in forecasting such as those cited in footnote 10.

Note: AG, CR, EN, GO and ID, are respectively, agricultural price index, crude oil price index, energy price index, gold price and industrial metal price index. EUR, AUD, BRP, CAD, and JPY respectively denote the dollar exchange rates of Euro, Australian dollar, British pound, Canadian dollar, and Japanese yen. A positive and statistically significant coefficient for the CW test implies that the larger model outperforms the smaller model, otherwise, it does not. The former has more parameters than the latter. The statistics reported are the regression coefficients obtained from the CW test equation while the values in parentheses () are the corresponding standard errors. ***, ** and * indicate statistical significance at 1%, 5% and 10%, respectively. Q and Y denote quarterly and annual frequencies respectively.

5.5.2. Controlling for other determinants of exchange rate

Under this scenario, we attempt to test the role of accounting for additional predictors of exchange rate in the forecast accuracy of the proposed commodity-based model. The choice of these predictors is motivated by the [Frenkel \(1976\)](#) and [Bilson \(1978\)](#) studies which argue for the role of interest rate differential and output differential, among other variables in the determination of exchange rate. The extended model clearly beats the baseline model that ignores some other important determinants of exchange rate both for the in-sample and out-of-sample forecasts (see [Table 16](#)).

6. Summary and concluding remarks

The probable relationship between commodity prices and exchange rates has thrown up an unending debate in the empirical literature with two strands of arguments explaining alternate causality between commodity prices and exchange rates. We contribute to this debate by focusing on the five tradable commodities, namely, agricultural price

¹⁰ Similar evidence is reported in studies focusing on different financial series, see [Narayan and Liu \(2015\)](#), [Narayan and Sharma \(2015\)](#), [Salisu et al. \(2016\)](#), [Salisu and Adeleke \(2016\)](#), and [Narayan et al. \(2018\)](#).

⁹ See also [Moosa and Burns \(2014 a,b,c\)](#).

index, crude oil price index, energy price index, industrial price index and price of gold. We also consider five most tradable currency pairs in the world involving the national currencies of six member countries of the OECD, namely, Australian dollar, British pound, Canadian dollar, Euro, Japanese yen, and the United States dollar, which in turn serves as the reference currency. The results of our preliminary analysis indicate the presence of persistence, endogeneity, serial correlation, and conditional heteroscedasticity effects in our predictors (that is, commodity prices), which in turn informs the choice of our estimator to account for these effects in the forecast model following [Lewellen \(2004\)](#) and [Westerlund and Narayan \(2012, 2015\)](#). We further extend the implementation of the estimator by accounting for the role of asymmetries and structural breaks.

We first establish a significant in-sample predictability of exchange rates using commodity prices irrespective of whether or not asymmetries are accounted for. This result is consistent with the findings of [Akram \(2004\)](#), [Ferraro et al. \(2015\)](#) and [Kohlscheen et al. \(2016\)](#), but is in contrast to the findings of [Mese and Rogoff \(1983\)](#), [Engel and West \(2005\)](#), and [Chen et al. \(2010\)](#). In contrast to these studies, we further evaluate the forecast performance of the commodity-based exchange rate forecast model (which happens to be our theoretical model) taking account of the possible role of asymmetries and structural breaks in the predictability of exchange rates. Contrary to the findings [Mese and Rogoff \(1983\)](#), [Chen et al. \(2010\)](#) and [Ferraro et al. \(2015\)](#), we find complementary results which reveal that accounting for asymmetries and structural breaks improve significantly both the in-sample and out-of-sample forecast performance of our proposed commodity-based. The results are robust to multiple currency pairs, commodity price indices, breaks and forecast horizons. However, in line with the studies of [Ferraro et al. \(2015\)](#), [Narayan and Liu \(2015\)](#), [Narayan and Sharma \(2015\)](#), [Salisu et al. \(2016\)](#), [Salisu and Adeleke \(2016\)](#), and [Narayan et al. \(2018\)](#), the forecast outcomes seem to diminish with lower data frequencies.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.resourpol.2019.03.006>.

References

- Apergis, N., Vouzavalis, G., 2018. Asymmetric pass through of oil prices to gasoline prices: evidence from a new country sample. *Energy Policy* 114, 519–528.
- Akram, Q.F., 2004. Oil prices and exchange rates: Norwegian evidence. *Econom. J.* 7 (2), 476–504.
- Ang, A., Bekaert, G., 2007. Return predictability: is it there? *Rev. Financ. Stud.* 20, 651–707.
- Ang, A., Bekaert, G., Wei, M., 2007. Do macro variables, asset markets, or surveys forecast inflation better? *J. Monet. Econ.* 54, 1163–1212.
- Arezki, R., Dumitrescu, E., Freytag, A., Quintyn, M., 2014. Commodity prices and exchange rate volatility: lessons from South Africa's capital account liberalization. *Emerg. Mark. Rev.* 19, 96–105.
- Bahmani-Oskooee, M., Saha, S., 2016. Do exchange rate changes have symmetric or asymmetric effects on stock prices. *Glob. Financ. J.* 31, 57–72.
- Bai, J., Perron, P., 2003. Computation and analysis of multiple structural change models. *J. Appl. Econom.* 18, 1–22.
- Bank for International Settlement (BIS), 2016. Foreign Exchange Turnover in April 2016. Triennial Central Bank Survey.
- Bannigidadmath, D., Narayan, P., 2015. Stock return predictability and determinants of predictability and profits. *Emerg. Mark. Rev.* 26, 153–173.
- Bilson, J., 1978. The monetary approach to the exchange rate – some empirical evidence. *Int. Monet. Fund Pap.* 25, 48–75.
- Bodart, V., Candelon, B., Carpentier, J.F., 2015. Real exchanges rates, commodity prices and structural factors in developing countries. *J. Int. Money Financ.* 51, 264–284.
- Bossaerts, P., Hillion, P., 1999. Implementing statistical criteria to select return forecasting models: what do we learn? *Rev. Financ. Stud.* 12, 405–428.
- Bouri, E., Gupta, R., Lahiani, A., Shabbaz, 2018. Testing for asymmetric nonlinear short- and long-run relationships between bitcoin, aggregate commodity and gold prices. *Resour. Pol.* <https://doi.org/10.1016/j.resourpol.2018.03.008>.
- Brennan, M., Xia, Y., 2005. Tay's as good as cay. *Financ. Res. Lett.* 2, 1–14.
- Broad, M., 2016. The Highs and Lows of the Pound. BBC News report. <https://www.bbc.com/news/business-36727177>.
- Butler, A., Grullon, G., Weston, J., 2005. Can managers forecast aggregate market return? *J. Financ.* 60, 963–986.
- Campbell, J.Y., 2008. Viewpoint: estimating the equity premium. *Can. J. Econ.* 41, 1–21.
- Chen, Y.C., Rogoff, K.S., Rossi, B., 2010. Can exchange rates forecast commodity prices? *Q. J. Econ.* 125 (3), 1145–1194.
- Chen, H., Liu, L., Wang, Y., Zhu, Y., 2016. Oil price shocks and US dollar exchange rates. *Energy* 112, 1036–1048.
- Chkili, W., Hammoudeh, S., Nguyen, D.K., 2014. Volatility forecasting and risk management for commodity markets in the presence of asymmetry and long memory. *Energy Econ.* 41, 1–18.
- Choudhri, E.U., Schembri, L.L., 2013. Productivity, commodity prices and the real exchange rate: the long-run behavior of the Canada–US exchange rate. *Int. Rev. Econ. Financ.* 29, 537–551.
- Clark, T.E., West, K.D., 2007. Approximately normal tests for equal predictive accuracy in nested models. *J. Econom.* 138, 291–311.
- Dauvin, M., 2014. Energy prices and the real exchange rate of commodity-exporting countries. *Int. Econom.* 137, 52–72.
- Devpura, N., Narayan, P.K., Sharma, S.S., 2018. Is stock return predictability time-varying? *J. Int. Financ. Mark. Inst. Money* 52, 152–172.
- Engel, C., West, K.D., 2005. Exchange rates and fundamentals. *J. Political Econ.* 113 (3), 485–517.
- Ferraro, D., Rogoff, K., Rossi, B., 2015. Can oil prices forecast exchange rates? An empirical analysis of the relationship between commodity prices and exchange rates. *J. Int. Money Financ.* 54, 116–141.
- Frankel, J.A., 2010. The Natural Resource Curse: A Survey. HKS Faculty Research Working Paper Series No. RWP10-005.
- Frenkel, J., 1976. A monetary approach to the exchange rate: doctrinal aspects and empirical evidence. *Scand. J. Econ.* 78, 255–276.
- Giacomini, R., Rossi, B., 2009. Detecting and predicting forecast breakdowns. *Rev. Econ. Stud.* 76 (2), 669–705.
- Goyal, A., Welch, I., 2003. Predicting the equity premium with dividend ratios. *Manag. Sci.* 49, 639–654.
- Harris, R.D., Shen, J., 2017. The intrinsic value of gold: an exchange rate-free price index. *J. Int. Money Financ.* 79, 203–217.
- Herrera, A.M., Lagalo, L.G., Wada, T., 2010. Oil Price Shocks and Industrial Production: Is the Relationship Linear? Mimeo, Wayne State U. 2010.
- Ignatieva, K., Ponomareva, N., 2016. Commodity currencies and commodity prices: modelling static and time-varying dependence. *Appl. Econ.* 49 (15), 1491–1512.
- Kilian, L., Vigfusson, R., 2011. Are the responses of the U.S. economy asymmetric in energy price increases and decreases? *Quant. Econ.* 2 (3), 419–453 2011.
- Kohlscheen, E., Avalos, F., Schimpf, A., 2016. When the Walk Is Not Random: Commodity Prices and Exchange Rates. Bank for Settlement (BIS) Working Papers Series No. 551.
- Lee, J., Strazicich, M.C., 2003. Minimum LM unit root test with two structural breaks. *Rev. Econ. Stat.* 85 (4), 1082–1089.
- Lewellen, J., 2004. Predicting returns with financial ratios. *J. Financ. Econ.* 74, 209–235.
- Lu, X., Li, J., Zhou, Y., Qian, Y., 2017. Cross-correlations between RMB exchange rate and international commodity markets. *Phys. Stat. Mech. Appl.* 486, 168–182.
- Lumsdaine, R., Papell, D., 1997. Multiple trend breaks and the unit root hypothesis. *Rev. Econ. Stat.* 79, 212–218.
- Makin, A.J., 2013. Commodity prices and the macroeconomy: an extended dependent economy approach. *J. Asian Econ.* 24, 80–88.
- Medina, J.P., Soto, C., 2016. Commodity prices and fiscal policy in a commodity exporting economy. *Econ. Modell.* 59, 335–351.
- Mese, R.A., Rogoff, 1983. Empirical exchange rate models of the seventies: do they fit out of sample? *J. Int. Econ.* 14, 3–24.
- Moosa, I., Burns, K., 2014a. The unbeatable random walk in exchange rate forecasting: reality or myth? *J. Macroecon.* 40, 69–81.
- Moosa, I., Burns, K., 2014b. Error correction modelling and dynamic specifications as a conduit to outperforming the random walk in exchange rate forecasting. *Appl. Econ.* 46 (25), 3107–3118.
- Moosa, I., Burns, K., 2014c. A reappraisal of the Mese–Rogoff puzzle. *Appl. Econ.* 46 (1), 30–40.
- Narayan, P.K., Bannigidadmath, D., 2015. Are Indian stock returns predictable? *J. Bank. Financ.* 58, 506–531.
- Narayan, P.K., Gupta, R., 2015. Has oil price predicted stock returns for over a century? *Energy Econ.* 48, 18–23.
- Narayan, P.K., Liu, R., 2015. A unit root model for trending time-series energy variables. *Energy Econ.* 50, 391–402.
- Narayan, P.K., Phan, D.H.B., Sharma, S.S., Westerlund, J., 2016a. Are Islamic stock returns predictable? A global perspective. *Pac. Basin Finance J.* 40 (A), 210–223.
- Narayan, P.K., Sharma, S.S., 2011. New evidence on oil price and firm returns. *J. Bank. Financ.* 35 (12), 3253–3262.
- Narayan, P.K., Sharma, S.S., 2015. Does data frequency matter for the impact of forward premium on spot exchange rate? *Int. Rev. Financ. Anal.* 39, 45–53.
- Narayan, P.K., Popp, S., 2010. A new unit root test with two structural breaks in level and slope at unknown time. *J. Appl. Stat.* 37 (9), 1425–1438.
- Narayan, P.K., Popp, S., 2013. Size and power properties of structural break unit root tests. *Appl. Econ.* 45 (6), 721–728.
- Narayan, P.K., Liu, R., Westerlund, J., 2016b. A GARCH model for testing market efficiency. *J. Int. Financ. Mark. Inst. Money* 41, 121–138.
- Narayan, P., Phan, D., Sharma, S., 2018. Does Islamic stock sensitivity to oil prices have economic significance? *Pac. Basin Finance J.* 53, 497–512.
- Nazlioglu, S., 2011. World oil and agricultural commodity prices: evidence from non-linear causality. *Energy Policy* 39 (5), 2935–2943.
- Paye, B.S., Timmermann, A., 2006. Instability of return prediction models. *J. Empir. Financ.* 13 (3), 274–315.

- Pesaran, M.H., Pick, A., Pranovich, M., 2013. Optimal forecasts in the presence of structural breaks. *J. Econom.* 177 (2), 134–152.
- Pesaran, M.H., Timmermann, A., 2007. Selection of estimation window in the presence of breaks. *J. Econom.* 137 (1), 134–161.
- Perron, P., 2006. Dealing with structural breaks. *Palgrave Handb. Econom.* 1, 278–352.
- Phan, D.H.B., Sharma, S.S., Narayan, P.K., 2015. Stock return forecasting: some new evidence. *Int. Rev. Financ. Anal.* 40, 38–51.
- Rafiq, S., Bloch, H., 2016. Explaining commodity prices through asymmetric oil shocks: evidence from nonlinear models. *Resour. Pol.* 50, 34–48.
- Rapach, D.E., Wohar, M.E., 2006. Structural breaks and predictive regression models of aggregate U.S. stock returns. *J. Financ. Econom.* 4 (2), 238–274.
- Rapach, D.E., Wohar, M.E., 2008. *Forecasting in the Presence of Structural Breaks and Model Uncertainty*, vol. 3 Emerald Group Publishing.
- Raza, N., Ali, S., Shahzad, S., Raza, S., 2018. Do commodities effectively hedge real estate risk? A multi-scale asymmetric DCC approach. *Resour. Pol.* <https://doi.org/10.1016/j.resourpol.2018.01.001>.
- Salisu, A.A., Adeleke, A.I., 2016. Further Application of Narayan and Liu (2015) unit root model for trending time series. *Econ. Modell.* 55, 305–314.
- Salisu, A.A., Swaray, R., Oloko, T.F., 2019. Improving the predictability of the oil-US stock nexus: the role of macroeconomic variables. *Econ. Modell.* 76 (C), 153–171.
- Salisu, A.A., Ndako, U.B., Oloko, T.F., Akanni, L.O., 2016. Unit root modeling for trending stock market series. *Borsa Istanbul Rev.* 16–2, 82–91.
- Salisu, A.A., Ademuyiwa, I., Isah, K., 2018a. Revisiting the forecasting accuracy of Phillips curve: the role of oil price. *Energy Econ.* 70 (C), 334–356.
- Salisu, A.A., Isah, K.O., 2018. Predicting US inflation: evidence from a new approach. *Econ. Modell.* 71, 134–158.
- Salisu, A.A., Oyewole, O.J., Fasanya, I.O., 2018b. Modelling return and volatility spillovers in global foreign exchange markets. *J. Inf. Optim. Sci.* 39 (7), 1417–1448.
- Shahzad, S., Hernandez, J., Al-Yahyaee, K., Jammazi, R., 2018. Asymmetric risk spillovers between oil and agricultural commodities. *Energy Policy* 118, 182–198.
- Shin, Y., Yu, B., Greenwood-Nimmo, M., 2014. Modelling asymmetric cointegration and dynamic multipliers in an ARDL Framework. In: Sickles, R., Horrace, W. (Eds.), *Festschrift in Honor of Peter Schmidt*. Springer, New York, NY, pp. 281–314.
- Smith, C.E., 1999. Exchange rate variation, commodity price variation and the implications for international trade. *J. Int. Money Financ.* 18 (3), 471–491.
- Trezzi, R., 2014. Exchange rates and commodity prices: granger causality in the time–frequency domain. *Appl. Econ. Lett.* 21 (3), 224–227.
- Ubilava, D., 2017. The ENSO effect and asymmetries in wheat price dynamics. *World Dev.* 96, 490–502.
- Valadkhani, A., Smyth, R., 2017. How do daily changes in oil prices affect US monthly industrial output? *Energy Econ.* 67, 83–90.
- Westerlund, J., Narayan, P.K., 2012. Does the choice of estimator matter when forecasting returns? *J. Bank. Financ.* 36, 2632–2640.
- Westerlund, J., Narayan, P.K., 2015. Testing for predictability in conditionally heteroscedasticity stock returns. *J. Financ. Econom.* 13, 342–375.
- Yang, L., Cai, X.J., Hamori, S., 2017. Does the crude oil price influence the exchange rates of oil-importing and oil-exporting countries differently? A wavelet coherence analysis. *Int. Rev. Econ. Financ.* 49, 536–547.